Transport and Shipping Operations

Course 3 — Transport, Warehousing and Port Operations · Module 7 · Teaching content for the Theory Builder

Review copy. Each lesson lists its blocks in entry order: enter each block's heading and body into the Theory Builder; enter practice questions as Lesson Checks; enter the module assessment via Configure assessment. Grey appendix sections are audit notes, not for entry.

Lesson 7.1 — Ocean Freight Operations and Vessel Schedules

Summary

This lesson examines the fundamental operations involved in transporting cargo by sea. Learners will explore how ocean freight services operate, the roles of shipping lines and vessel operators, and the procedures involved in planning shipments around vessel schedules. The lesson also introduces shipping routes, port rotations, cargo booking, vessel departure and arrival information, and the operational factors that may affect maritime transportation.

Learning objectives

- Define ocean freight and explain its importance in international trade.
- Identify the major participants involved in ocean freight operations.
- Distinguish between liner shipping and tramp shipping services.
- Explain how vessel schedules and port rotations influence shipment planning.
- Describe the basic procedures involved in booking cargo for ocean transportation.
- Interpret estimated and actual vessel departure and arrival information.
- Identify common factors that cause vessel delays and schedule changes.
- Explain the role of transshipment in international maritime transportation.
- Describe how shipping lines, freight forwarders, and port operators coordinate ocean freight shipments.

Key terms

Ocean Freight — The transportation of commercial goods by sea or other navigable waterways.

Liner Shipping — A maritime transportation service operating on established routes with published or planned schedules.

Tramp Shipping — Maritime transportation in which vessels are employed according to cargo availability and commercial arrangements rather than fixed scheduled routes.

Shipping Line — A company that operates or commercially manages maritime cargo transportation services.

Vessel Schedule — A planned timetable showing a vessel's expected port calls, departures, and arrivals.

Port Rotation — The planned sequence of ports visited by a vessel during a voyage or service.

Port of Loading (POL) — The port where cargo is loaded onto a vessel.

Port of Discharge (POD) — The port where cargo is unloaded from a vessel.

Estimated Time of Departure (ETD) — The anticipated date and time when a vessel is expected to depart.

Estimated Time of Arrival (ETA) — The anticipated date and time when a vessel is expected to arrive.

Actual Time of Departure (ATD) — The recorded date and time when a vessel actually departs.

Actual Time of Arrival (ATA) — The recorded date and time when a vessel actually arrives.

Transshipment — The transfer of cargo from one vessel or transport service to another during its journey to the final destination.

Feeder Vessel — A vessel used to transport cargo between smaller ports and larger hub ports or terminals.

Shipping Route — A maritime transportation path connecting ports or trading regions.

Teaching content

KEY CONCEPT

Ocean freight and its place in international trade

Course 3 moves the programme from foundations to operations, and Module 7 opens with the movement that carries most of the goods traded between continents: the sea. Lesson 3.2, Modes of Transport: Sea, Air, Road, Rail, and Multimodal, introduced sea transport as one mode among several. This lesson goes inside the mode and shows how ocean freight runs as a system: the services, the vessels, the schedules, and the cycle one container follows from the shipper's desk to the quay and across the ocean.

Ocean freight is the transportation of commercial goods by sea or other navigable waterways. The definition is wider than deep-sea voyages alone: coastal and river movements count as ocean freight too, wherever commercial goods travel by water.

Its importance to international trade rests on simple economics. A ship moves enormous quantities of cargo in one sailing, at a cost per unit of cargo that no other mode can match over long distances. For the heavy, bulky, and high-volume goods that flow between continents, grain, crude oil, clinker, vehicles, machinery, and the countless containers of finished goods, the sea is the only realistic carrier. For a trading nation such as Ghana, whose two seaports at Tema

and Takoradi are managed by the Ghana Ports and Harbours Authority, ocean freight is not one option among several; it is the physical basis of import and export trade.

That importance gives the industry its shape. Liner services sail on published schedules, terminals plan their work around vessel calls, forwarders plan shipments around sailing dates, and customers plan their businesses around arrival dates. Every one of those plans is the subject of this lesson.

KEY CONCEPT

The participants in ocean freight operations

No single organisation moves a container by sea. Ocean freight is performed by several participants whose roles interlock, and naming them is the first step towards understanding the coordination this lesson closes with. Lesson 2.2, Freight Forwarders, Carriers, Shipping Lines, and Agents, introduced the ecosystem; here the participants are seen through their ocean roles.

1. The shipping line. A shipping line is a company that operates or commercially manages maritime cargo transportation services. The line owns or charters vessels, publishes the schedules, sells the space, confirms bookings, issues bills of lading, and answers for the vessel's performance.

2. The shipper and the consignee. The shipper is the party whose goods travel, and the consignee is the party entitled to receive them at destination. They decide what moves, when, and on which service, and they supply the information and documents the rest of the network depends on.

3. The freight forwarder. Lesson 5.1, Role and Responsibilities of Freight Forwarders, defined the forwarder as the customer's agent in arranging transport. In ocean freight the forwarder usually books space with the line, arranges collection and delivery, prepares the shipping instructions, and manages the deadlines, the operational position this lesson's shipping cycle assumes.

4. Port and terminal operators. Lesson 2.3, Ports, Terminals, Warehouses, and Inland Depots, introduced ports and terminals. Terminal operators receive containers, stack them in the yard, and load and discharge vessels with cranes under plans agreed with the line; Module 9, Port and Terminal Operations, takes portside operations in depth.

Around these core parties work port authorities, customs administrations, and the ships' agents who represent the vessel in each port. The pattern to hold on to is dependence: each participant's plan is built from information the others supply.

KEY CONCEPT

Liner shipping and tramp shipping: two ways to hire the sea

Ocean freight services divide into two families, and the division decides how nearly every shipment is planned.

1. Liner shipping. Liner shipping is a maritime transportation service operating on established routes with published or planned schedules. A liner service sails the same route again and again, calls at the same ports in the same order, and departs on dates published weeks in advance; a weekly sailing is the common rhythm on the major container lanes. The line publishes its routes and its tariffs, and any shipper whose cargo fits the service can book space in it, one container at a time. Container shipping is the classic liner business, and the scheduled services this lesson follows are liner services.

2. Tramp shipping. Tramp shipping is maritime transportation in which vessels are employed according to cargo availability and commercial arrangements rather than fixed scheduled routes. A tramp vessel has no timetable and no published rotation; it is hired whole, normally under a contract of employment called a charter party, and it sails where the hirer's cargo requires, when that cargo is ready. The hirer does not buy a few containers of space; it takes the entire ship, and with it the commercial terms of the voyage.

3. Who uses which. The cargo decides. Manufacturers, retailers, and trading companies with containerised general cargo buy space on liner services, because their quantities are modest and their planning depends on regular sailings. Owners of bulk commodities, grain houses shipping harvests, mining houses moving ores, cement producers importing clinker, and energy companies moving crude oil and products, hire tramp vessels, because their cargoes fill whole ships and their routes follow the commodity from where it is found to where it is needed.

The two families fit in one sentence: liners sell space by the schedule; tramps sell the whole ship by the voyage.

KEY CONCEPT

Vessel types and the cargo each carries

Within the merchant fleet, five vessel types do most of the cargo work, and each exists because of a particular kind of cargo. This is the link Lesson 3.3, Cargo Types, Packaging, and Handling, makes visible from the cargo's side: the cargo decides the ship.

1. Container ships. These carry containerised cargo in standardised boxes stacked on deck and in the holds. They are the workhorses of liner shipping and the vessels this lesson's shipping cycle follows. Manufactured consumer goods, machinery, processed foods, and raw materials packed into containers all travel this way.

2. Dry bulk carriers. These carry unpackaged dry commodities poured directly into the hold: grain, ores, clinker, fertiliser, and similar cargoes of the kind Lesson 3.3 classes as bulk cargo. A bulk carrier's voyage is normally a tramp arrangement, one commodity, one hirer, one voyage.

3. Tankers. Tankers carry liquid bulk in the vessel's own tanks: crude oil, refined products, chemicals, and liquefied natural gas. Like the dry bulk trades, the tanker trades are overwhelmingly tramp and charter-based.

4. Roll-on/roll-off vessels, known as RoRo. RoRo vessels carry cargo that drives or rolls aboard under its own power or on wheels, cars, trucks, buses, and heavy plant, secured on vehicle decks; ramps replace cranes.

5. General or breakbulk vessels. The conventional multipurpose ships carry unitised or project cargo as separate pieces, pallets, crates, steel, and outsize project equipment too awkward for a container or a hold. These are the vessels of the breakbulk trade Lesson 3.3 introduced.

The mapping matters operationally. A forwarder arranging a movement chooses not only a route but a kind of service: containerised cargo goes to liner container services; a full cargo of rice goes to the tramp market. The rest of this lesson stays with the container trade, where most logistics careers begin.

KEY CONCEPT

The language of liner schedules: routes, rotations, and dates

A liner service is described by its route, its rotation, and its schedule, and the schedule speaks in four abbreviations.

A shipping route is a maritime transportation path connecting ports or trading regions, the trade lane at its largest scale, for example the route linking East Asia with West Africa.

A port rotation is the planned sequence of ports visited by a vessel during a voyage or service. Rotations are read as a list of ports in calling order, and the service repeats its rotation voyage after voyage. An illustrative rotation for a Far East service calling in West Africa reads: Shanghai, Singapore, Tema, Lome, Singapore, Shanghai. The string tells a shipper which ports the vessel calls, and in which order. The port of loading, POL, is the port where cargo is loaded onto a vessel; the port of discharge, POD, is the port where cargo is unloaded from a vessel. An exporter in Tema shipping to a buyer in Shanghai finds both ports in the rotation, in that order, and knows the service can carry the cargo.

A vessel schedule is a planned timetable showing a vessel's expected port calls, departures, and arrivals. Where the rotation says where a service goes, the schedule says when. These are what shippers plan against: a forwarder finds the sailing whose departure fits the cargo's readiness date, then plans the operation backwards from that sailing's deadlines.

The estimated time of departure, ETD, is the anticipated date and time when a vessel is expected to depart; the estimated time of arrival, ETA, is the anticipated date and time when a vessel is expected to arrive. Estimated carries the meaning: plans, not promises. The actual time of departure, ATD, is the recorded date and time when a vessel actually departs, and the actual time of arrival, ATA, is the recorded date and time when a vessel actually arrives, the vocabulary of tracking that Lesson 5.4, Shipment Tracking and Customer Communication, taught.

Interpretation is the skill. An ATD four hours late is noise; an ATD two days late moves every downstream date and puts each connection, a transhipment, an inland collection, a buyer's

warehouse slot, at risk until a revised ETA is published, and nothing is settled until an ATA is recorded. Disciplined operations compare actuals against estimates at every milestone and never quote an ETA as an appointment.

KEY CONCEPT

The shipping cycle: booking, confirmation, and equipment

The clearest way to teach liner operations is to follow one container. The cycle begins at the shipper's desk with the commercial process Lesson 5.2, Freight Quotations and Shipment Booking, taught in full: enquiry, quotation, acceptance, and then the booking.

The forwarder books space with the shipping line for a named sailing, and the line returns a booking confirmation stating, above all, the vessel and voyage, the estimated departure, the routing, and the cut-off deadlines for that sailing. Historically, a paper called a shipping order instructed the vessel to receive the booked cargo; today these workflows are almost entirely electronic, and the booking confirmation is the operative document. From the moment of confirmation, the shipment has a plan with dates attached, and deadlines.

The equipment then enters the cycle on one of two paths.

1. Full container load. For a container of its own, the line releases an empty container to the shipper or forwarder. The box travels to the shipper's premises, is stuffed with the cargo, and is sealed. The sealed box is now a single handling unit with a weight that must be established, the next block's subject.

2. Consolidated cargo. For cargo too small to justify a whole container, the goods are delivered loose to a container freight station, where the forwarder consolidates them with other customers' cargo into one shared container, as Lesson 5.3, Shipping Instructions and Cargo Consolidation, explained.

Either way, the clock is now running. The container, stuffed or to be built, must reach the terminal before a deadline the booking confirmation stated: the cargo cut-off.

KEY CONCEPT

The shipping cycle: verified gross mass and the cargo cut-off

Two obligations converge before a container may enter the terminal gate: a weight and a deadline.

The weight is a safety requirement under the International Convention for the Safety of Life at Sea, SOLAS. Verified gross mass, VGM, is the verified total weight of a packed container, and since 1 July 2016 the shipper must obtain and communicate it for every packed container before loading. The rules recognise two methods: Method 1 weighs the packed container on a scale, such as a weighbridge; Method 2 adds the container's tare weight to the weights of its contents and packaging, obtained with calibrated equipment and documented figures. The rule is

absolute: no container without a verified gross mass may be loaded aboard a vessel. The duty sits with the shipper, not with the line.

The deadline is the cargo cut-off: the date and time by which the packed container, with its VGM declared, must be delivered to the terminal, to the container yard, or to the container freight station for consolidated cargo. At the cut-off, the terminal's gate closes to that sailing's cargo.

Why does the gate close at all? Because loading a vessel is an engineered operation planned days in advance, and three bodies of data must be final before the plan can be built.

1. Stowage planning. The line and terminal decide where every container sits aboard, the stowage plan, and a plan built from unverified weights would endanger the vessel's stability.

2. Customs and manifest data. The export declaration and the vessel's manifest data for the cargo must be complete before departure.

3. VGM verification. The line and terminal must have each box's declared weight confirmed before the box can be worked.

The cut-off exists so those plans can be made from facts. A container that arrives after it is, operationally, a container the plan never accounted for.

KEY CONCEPT

The shipping cycle: documentation cut-off, loading, and departure

With the container inside the gate, a second clock is still running: the paperwork's.

1. The documentation cut-off. This is the deadline for finalising the shipping instruction, the consignee, cargo description, weights, marks, and container details from which the carrier prepares the bill of lading, as Lesson 5.3 taught. The instruction is the carrier's raw material for the transport document; after the cut-off, alterations are amendments to submitted details, and for the vessel's papers the details must stand as submitted. Commonly the documentation cut-off falls after the cargo cut-off, but the booking confirmation governs each sailing's actual dates.

2. Loading. Once the cargo cut-off has closed the gate, the terminal works the boxes. It arranges the sailing's containers in the yard in an order that serves the stowage plan, moves them to the quay, and the ship's cranes lift each box into its planned position aboard. The verified weights the shipper declared are load-bearing facts in the most literal sense: they determine where a box may sit and how it is secured.

3. Departure. The vessel sails, and the actual time of departure is recorded against the estimated one. The export declaration and the manifest data stand lodged; aboard, the carrier holds cargo for which it has issued, or is about to issue, its bill of lading, the document the final stage of the cycle turns to.

Then comes the part no shipper can manage: the voyage, lasting days or weeks according to the route, ending at the port of discharge.

A missed cut-off at the front of this cycle rolls the container to the next sailing, about a week on a weekly service, because the plans behind a sailing cannot absorb a late box; Lesson 5.2's rollover example showed the cost, and this lesson has now shown the reason.

KEY CONCEPT

The shipping cycle: arrival, discharge, and the bill of lading

At the far end of the voyage, the cycle reverses itself.

The line announces the vessel's approach through its local agents, and the actual time of arrival is recorded when the vessel reaches the port of discharge. The vessel berths and discharge begins: the cranes lift each container off, the terminal stacks them in the yard, and the cargo waits for its documents and its release.

The documents follow the pattern taught across Module 6. The bill of lading is issued by the carrier once the goods have been received and loaded aboard; Lesson 6.2, Bill of Lading and Air Waybill, taught its three functions as receipt for the goods, evidence of the contract of carriage, and document of title. The last of those gives the paper its mechanics: an original bill of lading controls the right to demand delivery, so a container is released only against an original bill, or against the carrier's authority under one of the paperless alternatives. Where the sale is paid through a bank, the originals travel through banking channels to the buyer. Where the parties trade on trust, the shipper may surrender the originals at origin for a telex release, and the carrier authorises electronic release at destination; or the parties may use a sea waybill, which names the consignee and requires no paper at all. Where a forwarder consolidated the cargo, the shipper holds the forwarder's house bill while the line's master bill covers the consolidated container.

On the importer's side, the goods are declared to customs at the port of discharge, in Ghana through ICUMS by a licensed clearing agent, as Lesson 6.1 described, and released against the transport document. The container is collected, returned empty to the line's depot, and the cycle that began at a desk closes at a gate.

KEY CONCEPT

Transshipment, feeder vessels, and hub ports

Not every container sails from its first port to its last on one vessel, and the networks that connect small ports to big ones have their own vocabulary.

Transshipment is the transfer of cargo from one vessel or transport service to another during its journey to the final destination. Lesson 2.3, Ports, Terminals, Warehouses, and Inland Depots, introduced the term alongside the facilities that make it possible; the concern here is why the networks are built this way and what transhipment costs a shipment.

Liner networks are built like wheels with rims and spokes. The largest vessels, the mainline services, sail the heavy trade lanes and call at a few great hub ports where enormous volumes can be exchanged. Smaller ports, and ports whose volumes cannot justify a mainline call, are connected to the hubs by feeder vessels: vessels used to transport cargo between smaller ports and larger hub ports or terminals. A container from a smaller port travels to its hub by feeder, joins the mainline vessel there, and reverses the process at the far end of the trade. The economics are compelling, one mainline sailing gathering the cargo of dozens of ports through a handful of hubs, and much of the world's cargo moves this way.

The cost is paid in time and in risk. A transshipped container's journey has two or more legs, and between the legs stands a connection: the box must reach the hub in time for the departure of its onward vessel. Each transfer adds handling, and each connection adds a point where a delay on one leg becomes a missed departure on the next, and at a hub the next departure may be days away. Planners therefore treat a transshipped routing's transit time as its direct-sailing equivalent plus a margin, and watch hub connections closely whenever schedules slip, a caution the worked examples return to.

KEY CONCEPT

Coordinating the network, and why schedules still slip

The cycle taught in this lesson shows three organisations holding one plan between them.

1. The shipping line owns the schedule. It publishes the rotations and dates, allocates vessel space, plans the stowage from the weights and instructions received, issues the booking confirmations and bills of lading, and announces every change when the plan moves.

2. The port and terminal operator owns the ground. It plans berths and yard space around the published schedule, receives containers against the cargo cut-off, and loads and discharges the vessel to the stowage plan.

3. The freight forwarder owns the customer's plan. It books against the schedule, arranges collection and gate-in before the cargo cut-off, submits the shipping instruction before the documentation cut-off, tracks the milestones Lesson 5.4 gave the vocabulary for, and keeps the customer informed.

Each party's plan is built from the others' information, which is why the deadlines exist and why accurate data, weights, instructions, and timings, is the currency of the relationship.

Yet the plan still slips, and the common causes are worth naming. Severe weather delays sailings and can close ports. Congestion at a busy port, too many vessels and too few berths or too little yard space, slows the work. Mechanical problems take vessels out of service for repair. A missed transhipment connection holds cargo at a hub until the next departure. And the network is chained: a vessel leaving one port late compresses the time available at every following port on its rotation, so one port's congestion becomes several ports' delay until the schedule recovers.

The professional response is not to promise perfection but to manage movement: monitor the actuals against the estimates, re-plan when the plan moves, and tell the customer before the customer asks. That is the operating posture of a working forwarder, and the closing habit of this lesson.

WORKED EXAMPLE

One container from Tema to Shanghai: the cycle in dates

A processor in Tema has sold 1,180 cartons of processed cocoa products to a distributor in Shanghai, packed into one 40-foot container. The forwarder is instructed to ship on a Far East liner service. All dates, times, and figures in this example are illustrative for teaching purposes.

Step 1: Booking and confirmation. On Friday the 6th the forwarder books space and receives a booking confirmation for the vessel on voyage 118W: routing Tema to Shanghai, estimated departure from Tema on Friday the 13th at 20:00, estimated arrival at Shanghai about twenty-eight days later, cargo cut-off Wednesday the 11th at 16:00, documentation cut-off Thursday the 12th at 12:00. The confirmation states the service rotation: Shanghai, Singapore, Tema, Lome, Singapore, Shanghai. The port of loading is therefore Tema and the port of discharge Shanghai.

Step 2: Equipment and stuffing. On Monday the 9th an empty 40-foot container is released to the exporter, trucked to the factory, stuffed with the 1,180 cartons on pallets, and sealed.

Step 3: Verified gross mass. On Tuesday the 10th the laden container crosses a weighbridge, Method 1, weighing the packed container. The verified gross mass is declared to the line that afternoon: the shipper's duty, and a condition of loading.

Step 4: Gate-in. On Wednesday the 11th at 11:00, five hours before the cargo cut-off, the container enters the terminal's container yard, its VGM already declared.

Step 5: Documentation. The shipping instruction, consignee, description, weights, and marks, was submitted complete on Tuesday, two days before the documentation cut-off. The carrier prepares the bill of lading from it.

Step 6: Loading, departure, voyage. The vessel works the terminal on Thursday night, and the line issues the bill of lading once the goods are loaded on board. Because the buyer is a long-standing account customer, the exporter surrenders the original bills at origin under a telex release, and the carrier authorises electronic release at destination. The estimated departure of Friday the 13th at 20:00 becomes an actual departure on Saturday the 14th at 02:20, a little over six hours late and unremarkable. The voyage takes about twenty-eight days.

Step 7: Arrival and discharge. The published ETA slips by one day; the actual time of arrival is recorded on the Monday, and the container is discharged and stacked in the yard at Shanghai. The consignee, cleared for electronic release, takes delivery as the named consignee without presenting paper.

What does this example demonstrate?

The sailing was a chain of deadlines and estimates, and it succeeded because every deadline was beaten by days, so a six-hour departure slip cost nothing.

Key lesson: book early, work backwards from the cut-offs, and read the actuals against the estimates, never instead of them.

WORKED EXAMPLE

Forty minutes late: a missed cargo cut-off and the customer conversation

A furniture manufacturer near Tema exports a 40-foot container of wooden furniture to a buyer in Jebel Ali on a weekly Middle East service. The booking confirmation states an estimated departure on Thursday, a cargo cut-off on Tuesday at 16:00, and a documentation cut-off on Wednesday at noon. All times and details are illustrative.

Step 1: The plan. The forwarder books the truck to gate-in on Monday, a full day before the cut-off, and the factory confirms that the stuffed, sealed container and its VGM declaration will be ready Monday morning.

Step 2: The slip. The factory's production runs late; the container is stuffed on Monday night instead, and the truck leaves for the terminal on Tuesday after lunch. On the highway, a breakdown ahead holds the truck for more than an hour. It reaches the terminal gate at 16:40, forty minutes after the cargo cut-off for that sailing.

Step 3: The rollover. The gate is closed to that sailing's cargo. The container cannot be worked into the vessel's stowage plan, which was finalised on the day's cut-off data. The line rolls the booking to the next sailing, seven days later, and issues revised cut-offs. The container gates in the following Wednesday, and the furniture sails a week late, arriving about a week after the date the sale anticipated.

Step 4: The customer conversation. The forwarder telephones the exporter the same afternoon, before the exporter hears of it from the buyer, and states the facts plainly: the container missed the cargo cut-off by forty minutes; the booking is rolled to next Thursday's sailing; the revised estimated arrival is a week later than promised; the additional costs are itemised and attributed; and the buyer's delivery commitment needs revising today. The call follows the discipline of Lesson 5.4, Shipment Tracking and Customer Communication: facts, impact, revised dates, and a written record the same day, not blame, and not silence.

Step 5: The prevention. Afterwards the parties change the plan's shape: gate-in is booked a full day before every cut-off, the factory's stuffing date is confirmed the day before the truck, and any risk to a deadline is escalated to the line before the cut-off passes, while options such as a later truck or an amended booking still exist.

What does this example demonstrate?

A sailing is planned from its cut-off data, so a late container is not a box that missed a truck; it is a box the vessel's plan never contained. Forty minutes became seven days.

Key lesson: the cargo cut-off is immovable. Plan to beat it by a day, and escalate before it passes, never after.

WORKED EXAMPLE

Shanghai to Basel: an arrival that needs a second leg

An engineering firm in Basel, Switzerland, a landlocked city far from any deep-sea port, buys machine components from a supplier in Shanghai. The routing the forwarder books is a study in transhipment. All times and figures are illustrative.

Step 1: The routing. The container sails on a mainline service from Shanghai to Rotterdam, a European hub, in about thirty days. From Rotterdam it continues on a scheduled Rhine barge service to Basel, with barge departures twice weekly, and the final kilometres to the factory are completed by truck. One sale, one container, three legs.

Step 2: The first leg. The container departs Shanghai on the mainline vessel. The firm's planner notes the published ETA at Rotterdam and, three days after it, the barge departure the forwarder has nominated for the onward leg.

Step 3: The missed connection. Congestion at the hub holds the mainline vessel, and its actual arrival at Rotterdam is recorded three days after the estimate, after the nominated barge has sailed. The next barge departure is four days later, and barges keep their own schedules and cut-offs just as deep-sea vessels do. The container waits in the hub's yard, and the connection it missed cannot be recovered; it can only be replaced by the next one.

Step 4: The second leg and arrival. The container is loaded aboard the next barge, sails up the Rhine, and reaches Basel, where a truck completes the delivery to the factory. The door-to-door journey has taken about a week longer than the ocean ETA alone suggested.

Step 5: What the second leg adds. The firm learns the arithmetic of hub routings: total transit equals the mainline leg, plus the wait for the connection, plus the inland leg. Each connection is a fresh chance for one leg's delay to become the next leg's absence, so a transshipped routing needs a planning margin that a direct routing does not.

What does this example demonstrate?

Transshipment is not a delay in itself; it is how landlocked and smaller destinations reach the world's mainlines. But every transfer point converts upstream delay into downstream waiting unless the plan carries a margin.

Key lesson: plan hub-to-door transit from the final destination, not from the first port of discharge, and treat every connection as a date that can be missed.

IMPORTANT NOTE

The schedule is a plan, not a promise

Every date a shipping line publishes, the ETD, the ETA, the port calls on the rotation, is a plan, and the plan is exposed to weather, congestion, mechanical trouble, and the accumulated slippage of every port before yours. Three habits follow.

First, read estimates as estimates. A booking confirmation's dates state the service's intention, not a guaranteed arrival time. Plans that genuinely matter, a buyer's launch, a factory's production run, a vessel connection, need a buffer between the published ETA and the date the goods are truly required.

Second, expect delay to cascade. A vessel that leaves one port half a day late rarely recovers by the next: the rotation's remaining calls compress, terminal windows shift, and the delay can travel with the ship for days until the schedule's built-in slack absorbs it. This is why an ATD recorded at a port thousands of kilometres away is still the importer's information; it moves the ETA.

Third, treat reliability as variable. Some services and trades run to time more dependably than others, and the same service can differ by season and by port. That judgement is qualitative, and it belongs in every routing decision: a tight connection on a lane known for congestion is a risk bought knowingly or not at all.

None of this excuses poor communication. The forwarder's duty, taught in Lesson 5.4, is to pass schedule changes on the day they are known, with a revised estimate and its consequences, because a customer can manage a delay it is told about, and cannot manage one it discovers.

IMPORTANT NOTE

Verified gross mass is the shipper's obligation

The VGM rule deserves its own warning, because the most dangerous habit in container shipping is guessing a weight.

The obligation is the shipper's. Under SOLAS, the shipper must obtain and communicate the verified gross mass of every packed container, by weighing the packed container or by adding the container's tare to contents weights obtained with calibrated equipment. The carrier does not weigh your box for you, and an estimated figure from a packing list is not a verified gross mass.

The reason is physical. A container ship's stability, its stacking plan, and the lashings that hold thousands of boxes through storms are all calculated from the declared weights. A container declared at 12 tonnes that actually weighs 19 tonnes sits where no 19-tonne box was meant to sit. Mis-declared weights can collapse stacks, injure crew, and force a vessel to re-stow mid-voyage. An inaccurate VGM is therefore not a paperwork error; it endangers the vessel, the crew, and every other shipper's cargo.

The consequence is commercial as well, and it lands on the shipper's own box: a container with no verified gross mass, or with a declaration the line cannot accept, will not be loaded. It waits ashore, missing its sailing and usually its customer's date, until the weight is properly declared.

The professional habit is short: weigh or calculate honestly, declare early, and treat Method 2 figures as measured facts from calibrated equipment, never as arithmetic built on estimates. The weight you declare is a promise the ship is loaded on.

IMPORTANT NOTE

Never assume a cut-off: the booking confirmation governs

Cut-offs are the most dangerous numbers in the cycle to carry in your head from a previous shipment, because they are never general; they are specific.

They differ per port: the same line's cut-offs at Tema and at its next port of call differ, because different terminals work to different rhythms. They differ per line: two lines calling the same port on the same day can close their cargo gates hours apart. They differ per cargo and equipment: consolidated cargo built at a container freight station, refrigerated cargo needing inspection before shipment, and out-of-gauge cargo all face their own arrangements and often earlier deadlines. And they move week to week: a service's cut-offs shift with the rotation's timing, so last Thursday's cut-off tells you nothing certain about this Thursday's.

The only source that matters is the booking confirmation for the sailing actually booked: the vessel, the voyage, the cargo cut-off, the documentation cut-off. Reading it on the day it arrives, and planning backwards from its dates, is what turns the shipping cycle from a scramble into a schedule.

The professional habits are short. Check the confirmation against the operation the day it is received. Put the cut-offs into the plan before the truck is booked, not after. Chase the shipping instruction days before the documentation cut-off, not hours before. And when a deadline is at risk, call the line before it passes, because options exist before a cut-off that no longer exist after it.

Practice questions (Lesson Checks)

Practice Question 1

A grain trader has 25,000 tonnes of maize to move between two ports, with timing flexible within a month. A furniture exporter has one 40-foot container of goods to move on a trade lane served by a weekly container service.

Which statement best describes the service each should use?

1. Both should use liner services, because all ocean cargo travels on published schedules.

2. The grain trader should hire a tramp vessel for the whole cargo under a charter arrangement, while the furniture exporter should book space with a liner service sailing on a published schedule. ✓

3. Both should hire tramp vessels, because tramp shipping is always cheaper than liner shipping.

4. The grain trader should book space on the weekly container service, while the furniture exporter should charter a whole vessel for its single container.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The cargo decides the service. Tramp shipping employs vessels according to cargo availability and commercial arrangements rather than fixed schedules: a trader with 25,000 tonnes of maize fills a whole ship, so the commercial answer is to hire the vessel entirely, normally under a charter party, and send it where the cargo requires. Liner shipping operates on established routes with published or planned schedules and sells space by the container: an exporter with one 40-foot container cannot fill a ship and needs the regular sailings of a scheduled service to plan against.

Answer 1 fails because tramp cargo does not travel on published schedules at all; the two families exist precisely because cargoes differ. Answer 3 fails because tramp shipping is not always cheaper; cost depends on the cargo, and a single container chartered alone would be absurdly expensive. Answer 4 reverses the logic entirely: a bulk cargo of 25,000 tonnes cannot travel as container space, and one container cannot justify a whole vessel.

The takeaway: liners sell space by the schedule and suit containerised general cargo; tramps sell the whole ship by the voyage and suit bulk commodities.

Practice Question 2

A forwarder holds a booking confirmation stating an estimated departure on Thursday, a cargo cut-off on Tuesday at 16:00, and a documentation cut-off on Wednesday at 12:00. The container will be stuffed only on Wednesday morning, and the weighbridge appointment for its verified gross mass is on Wednesday afternoon.

Which statement best describes the shipment's position?

1. The shipment is safe, because the vessel does not depart until Thursday and the container only needs to reach the terminal before the vessel sails.

2. The plan fails the cargo cut-off: the packed container with its VGM declared must be at the terminal by Tuesday at 16:00, so a Wednesday gate-in cannot load, and the operation must be expedited or moved to the next sailing. ✓

3. The position is acceptable, because the verified gross mass can be waived when the packing list already states the gross weight of the goods.

4. The position is acceptable, because the documentation cut-off is the only binding deadline, and the shipping instructions can still be submitted on Wednesday morning.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The cargo cut-off is the deadline by which the packed container, with its verified gross mass declared, must be delivered to the terminal, because the line and terminal finalise the stowage plan, the manifest data, and the weight records from the cut-off position. A container stuffed on Wednesday and weighed on Wednesday afternoon cannot gate in by Tuesday at 16:00, so it cannot load on Thursday's vessel; the choices are to expedite the operation against a revised arrangement with the line or to accept a rollover to the next sailing.

Answer 1 fails because the deadline that matters is the cut-off, not the departure; the gate closes a day or more before the vessel sails. Answer 3 fails because verified gross mass is a shipper's obligation under SOLAS and is never waived; a packing list weight is an estimate, not a verification. Answer 4 fails because there are two separate deadlines and both bind: meeting the documentation cut-off while missing the cargo cut-off still leaves the container ashore.

The takeaway: plan backwards from the cargo cut-off, not from the departure date, and treat the VGM declaration as part of meeting it.

Practice Question 3

An importer's booking confirmation showed an ETD on the 3rd and an ETA on the 28th, with a transhipment at a hub port. The line has now recorded the ATD as the 5th. The customer asks whether the goods will still arrive on the 28th.

Which response reflects correct professional practice?

1. Yes, because the ETA stated on the booking confirmation is a commitment that does not change after the booking is made.

2. No, but the exact arrival date is already known, because once the ATD has been recorded the ATA is determined with it.

3. The departure slipped two days, so the ETA must be re-assessed and the customer given a revised estimate, especially where the transhipment connection can be missed; only the ATA will confirm the actual arrival. ✓

4. The delay entitles the importer to have the cargo transferred to a faster vessel at no additional cost.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The ETD and the ETA are estimates, and the ATD is a recorded fact showing the plan has already slipped by two days. Every downstream date timed against the old estimate is now questionable, and on a transshipped routing the risk is sharper still: the container must reach the hub in time for its onward vessel, so a late arrival can mean a missed connection and a wait of days for the next departure. Correct practice is to re-assess the ETA in the light of the ATD, obtain the line's revised estimate, and inform the customer promptly, exactly the tracking and communication discipline Lesson 5.4 taught. The actual time of arrival is recorded only when the vessel actually arrives.

Answer 1 fails because an ETA is an estimate, not a promise. Answer 2 fails because the ATD does not determine the ATA; delays, congestion, and connections stand between the two. Answer 4 invents an entitlement that does not exist; the remedy is honest re-planning, not a guaranteed transfer.

The takeaway: treat ETD and ETA as plans, watch the actuals, re-assess connections after any slip, and communicate the revised estimate the day it is known.

Fact-check appendix (audit notes — not for entry)

Definitions of ocean freight, liner shipping, tramp shipping, shipping line, vessel schedule, port rotation, port of loading (POL), port of discharge (POD), ETD, ETA, ATD, ATA, transhipment, feeder vessel, and shipping route follow the lesson brief's key terms. — source: lesson brief 7.1, Key Terms (curriculum database).

Liner services run published routes and schedules at published tariffs (container shipping); tramp vessels go where cargo demands under charter parties. — source: research sheet trade-transport-standards.md, section 7 (ocean freight operations, liner vs tramp).

Principal vessel types: container ships; dry bulk carriers (grain, ores, clinker); tankers (crude/products/chemicals/LNG); RoRo (roll-on/roll-off for vehicles); general/breakbulk (unitised or project cargo on conventional ships). — source: research sheet trade-transport-standards.md, section 7.

Booking flow: cargo is booked with the line (or via a forwarder); the line confirms with a booking confirmation; historically a shipping order instructed the ship to receive cargo, and today the workflows are mostly electronic. — source: research sheet trade-transport-standards.md, section 7 (the sheet marks current shipping-order usage as medium confidence, so it is taught as history).

Cut-offs before each sailing: the cargo/VGM cut-off is when the packed, VGM-declared container must be at the terminal (container yard) or CFS; the documentation (shipping instruction) cut-off is the deadline for the bill of lading details; missing a cut-off usually means rollover to the next vessel. — source: research sheet trade-transport-standards.md, section 7.

The operational reasons cut-offs exist, stowage planning, complete export/manifest data before departure, and VGM verification, are general business knowledge built on the sheet's cut-off and VGM entries. — source: research sheet trade-transport-standards.md, section 7, plus general business knowledge.

Verified gross mass (VGM): SOLAS Regulation VI/2, mandatory since 1 July 2016; the shipper must obtain and communicate the verified gross mass of every packed container, by Method 1 (weighing the packed container) or Method 2 (tare plus contents using calibrated equipment, per IMO Guidelines MSC.1/Circ.1475); no container without VGM may be loaded. — source: research sheet trade-transport-standards.md, section 7.

Ocean bill of lading functions: receipt for the goods, evidence of the contract of carriage, and document of title; the sea waybill is the non-negotiable alternative releasing cargo to the named consignee without paper; telex (express) release means the shipper surrenders the originals at origin and the carrier authorises electronic release at destination; the master B/L is issued by the ocean carrier and the house B/L by the freight forwarder. — source: research sheet trade-transport-standards.md, section 4 (transport documents).

Weekly sailing frequency on major container lanes is the industry's common rhythm. — source: general business knowledge, as taught in the lesson 5.2 draft.

Most goods traded between continents travel by sea. — source: general business knowledge, stated qualitatively without statistics.

Ghana's two seaports, Tema and Takoradi, are managed by the Ghana Ports and Harbours Authority (GPHA). — source: research sheet ghana-trade-customs.md, section 4 and its SAFE PHRASINGS.

In Ghana the customs declaration for an imported consignment is lodged electronically in ICUMS (the Integrated Customs Management System), normally by a licensed clearing agent acting for the importer. — source: research sheet ghana-trade-customs.md, sections 1 and 6, as taught in the lesson 6.1 draft.

Common causes of vessel delay and schedule change named in the lesson (weather, port congestion, berth and yard constraints, mechanical problems, missed transhipment connections, and cascading delays along a rotation) are taught qualitatively; no reliability percentages or index figures are used. — source: general business knowledge; the DO NOT CLAIM list forbids current reliability statistics and freight-index levels.

Liner pricing concepts (published tariffs, freight-rate components, general rate increases) are acknowledged only in passing; their detailed treatment belongs to Lesson 7.4, Shipping Charges, Freight Rates, and Surcharges, per the anti-repetition rule. — source: WRITING_SPEC.md section 7; research sheet trade-transport-standards.md, section 7.

The Rotterdam–Basel Rhine barge routing, the Tema–Shanghai rotation, all vessel and voyage numbers, dates, times, transit times, and quantities in the worked examples are illustrative for teaching purposes. — source: WRITING_SPEC.md fact-safety rule 5.1.

Lesson 7.2 — Air Cargo Operations and Cargo Acceptance

Summary

This lesson introduces the operational procedures involved in transporting goods by air. Learners will examine the roles of airlines, air cargo terminals, freight forwarders, and ground

handling agents in preparing, accepting, handling, and transporting shipments. The lesson also explains cargo acceptance requirements, shipment booking, air waybill documentation, security screening, and the handling of cargo requiring special transportation conditions.

Learning objectives

- Define air cargo operations and explain their role in international logistics.
- Identify the major participants involved in air freight transportation.
- Describe the basic procedures for arranging an air cargo shipment.
- Explain the requirements for cargo acceptance by airlines and handling agents.
- Identify the purpose of air waybills and other air cargo documentation.
- Describe common air cargo screening and security procedures.
- Explain the importance of correct packaging, marking, and labelling for air shipments.
- Identify cargo categories requiring special handling or acceptance procedures.
- Describe how air cargo is handled from origin acceptance to destination delivery.
- Compare the operational characteristics of air freight and ocean freight transportation.

Key terms

Air Cargo — Goods transported by aircraft for commercial or logistical purposes.

Air Freight — The transportation of goods by aircraft.

Air Cargo Terminal — A facility at or near an airport used for receiving, handling, storing, and processing air cargo.

Ground Handling Agent (GHA) — A company providing ground-based services to airlines, including specified cargo handling activities.

Cargo Acceptance — The process of examining and receiving cargo to determine whether it satisfies applicable transportation requirements.

Air Waybill (AWB) — A non-negotiable air transport document that provides evidence of the contract of carriage and acknowledges receipt of cargo.

Master Air Waybill (MAWB) — An air waybill issued by an airline or its authorised agent covering cargo accepted under the airline's contract of carriage.

House Air Waybill (HAWB) — An air transport document issued by a freight forwarder to a shipper for an individual consignment.

Air Cargo Screening — Security procedures used to examine cargo for prohibited articles or potential threats before transportation.

Unit Load Device (ULD) — An aircraft pallet or container used to consolidate and secure cargo, baggage, or mail for air transportation.

Chargeable Weight — The weight used to calculate air freight charges, determined according to applicable actual-weight and volumetric-weight rules.

Special Cargo — Cargo requiring additional handling, documentation, packaging, or transportation precautions because of its characteristics.

Perishable Cargo — Goods that may deteriorate unless transported and stored under suitable environmental conditions.

Dangerous Goods — Substances or articles that present hazards during transportation and are subject to applicable safety regulations.

Cargo Manifest — A document or electronic record listing cargo carried on a particular transport service.

Teaching content

KEY CONCEPT

The air freight product and why shippers choose it

Air cargo is goods transported by aircraft for commercial or logistical purposes, and air freight is the transportation of goods by aircraft. Behind those definitions sits a distinct product with its own participants, documents, and operating discipline. Lesson 7.1, Ocean Freight Operations and Vessel Schedules, examined the sea leg of world trade; this lesson turns to the air above the same lanes and, centrally, to cargo acceptance, the gatekeeping process every air consignment must pass before it flies.

Businesses choose air freight when the value of speed outweighs its cost.

1. Speed. An aircraft crosses in hours what a vessel crosses in weeks. For urgent shipments the difference is not convenience but the point of the transaction: a production line stopped for a missing part, a customer waiting for a sample, a market window for fresh produce.

2. Security. Air cargo passes through controlled airport environments, screening procedures, and closely supervised handling. Exposure time is short, and the cargo's whereabouts are known at every stage.

3. Lower in-transit exposure. Because goods spend hours rather than weeks in transit, less capital is tied up in inventory on the move, insurance exposure over the journey is shorter, and goods reach the market sooner. These savings partly offset air freight's higher price per consignment.

What suits air? Perishable produce, pharmaceuticals, electronics and other high-value goods, urgent spare parts, and samples and documents are the classic air commodities: cargo whose value or vulnerability makes time the decisive factor.

Air is also an airport-to-airport product. An aircraft cannot call at a factory gate or a shop, so every air movement is completed by road: collection to the origin airport and delivery from the destination airport. Road feeder services, truck movements operating as part of the air network under air cargo documents, extend the air product to points no aircraft serves directly, so the waybill may name airports at either end while the goods travel the first or last kilometres by road.

KEY CONCEPT

The participants in the air cargo chain

An air consignment passes through more hands than a new exporter expects. Lesson 2.2 introduced the ecosystem of intermediaries; in air freight the main participants are these.

1. The shipper. The exporter sending the goods. The shipper prepares the consignment, states its contents, weight, and dimensions, signs the transport document, and bears responsibility for the accuracy of that declaration.

2. The freight forwarder. The organiser of the shipment, introduced in Lesson 5.1, Role and Responsibilities of Freight Forwarders. In air freight the forwarder books space, prepares the documentation, delivers the goods to the terminal within the acceptance deadlines, and frequently consolidates several customers' consignments into one airline booking.

3. The airline. The carrier operating the aircraft and selling cargo capacity, either directly to shippers with substantial volumes or through forwarders. Some aircraft carry cargo in the holds of passenger services alongside baggage; dedicated freighter aircraft carry cargo as their main payload.

4. The air cargo terminal. The facility at or near the airport used for receiving, handling, storing, and processing air cargo. Terminals are where cargo is checked, screened, stored, and prepared for the aircraft, and where arriving cargo is broken down for collection.

5. The ground handling agent. A company providing ground-based services to airlines, including specified cargo handling activities. At the terminal, the ground handling agent works on the airline's behalf: it checks and accepts cargo, builds unit loads, loads and unloads aircraft, and manages the arrival process.

6. Customs and other authorities. Goods crossing borders pass export and import formalities, and some commodities carry additional health, plant-health, or security controls depending on the countries involved.

The practical picture is simple: the forwarder is the face of the operation, the airline provides the flying, the ground handling agent executes the terminal work, and the authorities control what may cross the border. Acceptance is where all these parties' requirements converge on one consignment.

KEY CONCEPT

Booking an air shipment and confirming space

Every air movement begins with a booking, the reservation of cargo space on a named flight. The procedure runs in a familiar sequence.

1. Enquiry and quotation. The shipper, usually through a forwarder, describes the consignment: what the goods are, how many packages, the weight and dimensions of each package, any special handling needs, when the goods will be ready, and the destination. The forwarder or airline responds with a service proposal and a price calculated on the chargeable weight, taught in the next concept.

2. Booking request. The forwarder asks the airline for space on a suitable flight. Capacity is finite: a passenger hold fills alongside baggage, a freighter's payload and volume are fixed, and peak seasons leave space tight.

3. Booking confirmation. The airline confirms the space, the flight, and the acceptance details. Only at confirmation is the shipment's place on the aircraft assured.

4. Preparation. The shipper packs, marks, and labels the goods and assembles the documents, and the forwarder prepares the air waybill data and supporting paperwork, such as health or phytosanitary certificates where required.

5. Delivery within the acceptance cut-off. The consignment must reach the airline's cargo terminal before the acceptance deadline for the booked flight.

The details given at booking matter more than beginners assume: the airline plans the load, the space, and the handling from exactly those figures. If the goods delivered differ from the figures booked, the discrepancy surfaces at acceptance, and the shipment may be refused or held until the booking is corrected.

KEY CONCEPT

Chargeable weight: why air freight weighs space as well as kilograms

Ocean freight, as Lesson 7.1 showed, is charged mainly by container or by the tonne; air freight is charged by weight, but by a special kind of weight. Chargeable weight is the weight used to calculate air freight charges, determined according to applicable actual-weight and volumetric-weight rules. In everyday terms: the carrier takes the greater of the actual gross weight and the volumetric weight, and charges that.

Volumetric weight is a conversion of the space a consignment occupies into equivalent kilograms. Under the IATA standard formula, volumetric weight in kilograms equals the length multiplied by the width multiplied by the height, in centimetres, divided by 6,000. A carton of 60 by 50 by 40 centimetres therefore has a volumetric weight of 120,000 divided by 6,000, which is

20 kilograms, regardless of what it actually weighs. Some carriers and contracts apply different divisors, so the applicable rule should always be confirmed for the trade lane concerned.

Why does the industry charge this way? Because an aircraft sells two scarce resources at once: payload and space. The hold has a maximum weight it may lift and a maximum volume it may fill, and whichever runs out first is the resource the cargo truly consumes.

1. Dense cargo exhausts the weight first. A carton of machine parts may reach the payload limit while occupying only a slice of the hold. Charging it by actual weight is fair to the airline.

2. Light, bulky cargo exhausts the space first. Polystyrene packaging, lampshades, and plastic housewares fill the hold long before the aircraft is heavy. If the airline charged only actual weight, it would be selling floor-to-ceiling air for the price of a few kilograms.

The chargeable weight rule makes every consignment pay for whichever resource it uses more of. The arithmetic is worked through in a later worked example; the habit is simple: measure every package as well as weighing it, because for light goods the tape measure, not the scales, decides the freight.

KEY CONCEPT

The air cargo terminal and the acceptance cut-off

Between the shipper's premises and the aircraft stands the air cargo terminal, the facility at or near an airport used for receiving, handling, storing, and processing air cargo. The terminal is the air-freight equivalent of the port container terminal taught alongside Lesson 7.1, compressed into a far shorter timescale. Its core functions run in sequence.

1. Receiving and acceptance checking. Cargo is delivered to the terminal, where its documents and physical condition are checked before the airline accepts it.

2. Security screening. Consignments are screened or arrive under approved security status, as explained later in this lesson.

3. Temporary storage. Cargo waits for its flight in secure storage, with cold rooms for perishables and segregated areas for special cargo.

4. Build-up. Loose packages are assembled into unit load devices, the aircraft pallets and containers described later in this lesson.

5. Breakdown. On arrival, the sequence runs in reverse: unit loads are broken down, cargo is checked, notified, and released to the consignee after customs formalities.

The terminal's operating rhythm is set by acceptance cut-off times: the deadlines by which a booked consignment must be delivered, checked, and accepted before its flight. Acceptance is not a single stamp but a process of document checks, physical checks, screening, and build-up, each needing time before loading. Cut-offs therefore close well before departure, and they vary

by airline, airport, flight type, and commodity; a shipper works to the deadline given for each booking.

The consequence is the one beginners find hardest to believe: cargo can miss a flight that has not yet departed. An aircraft will leave with hold space empty rather than take a consignment that arrived after acceptance closed, because the paperwork and the load plan close first. Worked Example 1 shows the clock in operation.

KEY CONCEPT

Cargo acceptance: the gatekeeping function

Cargo acceptance is the process of examining and receiving cargo to determine whether it satisfies applicable transportation requirements. It is the central operational function of this lesson, because acceptance is where the air industry protects itself: the safety of the aircraft, the security of the flight, the accuracy of the documents, and the flow of every other consignment depend on refusing what does not meet the requirements. The airline or its ground handling agent performs acceptance at the terminal, and it has two sides, the documents and the physical goods, checked together.

1. The document check. Acceptance staff compare the paperwork against the booking: the air waybill details must match the parties, the routing, the number of pieces, the weight, and the description of the goods. The supporting attachments must be present where the goods require them: a dangerous goods declaration for regulated hazardous articles, phytosanitary certificates for plants and plant products, health certificates for products of animal origin, and any other required permits. A document that disagrees with the booking, or a missing attachment, stops the process until it is resolved.

2. The physical check. The consignment itself is examined. Pieces are counted against the document. The consignment is weighed and measured against the stated weight and dimensions, the figures the booking and chargeable weight were built on. Packaging is inspected for condition and suitability: damage, leaks, or packing inadequate for the unit-load handling and stacking of air carriage are grounds for refusal. Finally, marking and labelling are checked, including hazard labels for any dangerous goods.

Acceptance ends in one of two outcomes. The consignment is accepted, the acceptance is recorded, and the cargo proceeds to screening and build-up; or a check fails and the cargo is refused until the fault is corrected. The decisive principle is that the goods, the documents, and the booking must agree. Worked Example 3 shows a rejection, and why the rejection is the system working rather than failing.

KEY CONCEPT

Packaging, marking, and labelling for air carriage

Lesson 3.3, Cargo Types, Packaging, and Handling, treated packaging as the protection of goods in transit. Air carriage adds three further pressures, and each shapes what the acceptance desk will and will not take.

1. Handling and stacking. Air cargo is built into unit loads, stacked inside aircraft containers and on pallets, and handled by equipment at every point of the journey. Packaging must be strong enough for stacking and unit-load handling even though air shipments are lighter than their sea counterparts.

2. Weight economy. Every kilogram of packaging can become chargeable kilograms, because the charge follows the greater of actual and volumetric weight. Air packaging is therefore designed to be light as well as strong, a balance shippers of dense goods in particular learn to appreciate.

3. Identification under pressure. Cargo at a terminal changes hands quickly among people who have never seen the consignment before. The packages must speak for themselves.

That last pressure explains the distinction between marking and labelling. Marking is the identification applied to the packages: the marks and package numbers, the shipper's and consignee's details, and the handling information that ties each physical piece to the air waybill. Markings must agree with the transport document, because at acceptance the two are compared. Labelling is the application of standard labels instructing the chain how to treat the package: handling labels indicate treatment such as orientation or fragility, and hazard labels declare and identify regulated dangerous goods, accompanying the shipper's dangerous goods declaration. A perishable consignment may carry special-handling labels so it is routed to cold storage.

The operational rule is blunt: packages that arrive unmarked, mislabelled, or in packaging unsuitable for air carriage fail the physical acceptance check, and the cargo does not fly until the fault is corrected. Labels are not decoration; they are the instructions the whole terminal acts on.

KEY CONCEPT

Security screening and the known-consignor concept

Air cargo is security cargo. It travels in aircraft alongside people, and the security of the flight depends on no prohibited article entering the hold. Air cargo screening is the security procedure used to examine cargo for prohibited articles or potential threats before transportation, and no cargo may be loaded onto an aircraft unless its security status is established.

The concept works through two broad routes, and the lesson teaches them generically, because every country implements the principle through its own national regime.

1. Cargo from approved trusted shippers. Security regimes around the world allow shippers whose premises, personnel, and procedures have been assessed and approved to send cargo whose security is established at origin, on the strength of the shipper's own approved supply-chain controls. Such shippers are commonly described as known consignors; their consignments can enter the air side without further screening, though the approval is conditional on maintaining the required procedures.

2. Screened cargo. Cargo that does not arrive under an approved shipper's security status must be examined before acceptance into the air side. In practice the consignment is screened at or before the terminal by recognised means, which include scanning equipment and physical examination, and its screened status is recorded.

The security status travels with the consignment as data and paperwork, exactly as the transport document does: the terminal can see, for every piece, whether it is screened or covered by an approved shipper's status. Cargo whose status cannot be established is not loaded; it waits until screening is completed, which in practice can mean a missed flight.

For the shipper, two conclusions follow: security status is part of the acceptance picture, so the time before a cut-off must cover it; and trusted-shipper approval speeds a regular exporter's consignments through the terminal, but is earned through sustained compliant procedures, not claimed at the desk.

KEY CONCEPT

Special cargo, perishables, and dangerous goods acceptance

Special cargo is cargo requiring additional handling, documentation, packaging, or transportation precautions because of its characteristics. The category exists because the acceptance procedures taught so far, documents, physical checks, screening, and build-up, are the standard path, and some commodities cannot safely travel that path without additions.

1. Perishable cargo. Goods that may deteriorate unless transported and stored under suitable environmental conditions, such as fresh produce, flowers, and certain foods. Perishables demand a cold chain, the unbroken sequence of temperature-controlled handling from packhouse to aircraft: cold rooms at the terminal, minimum waiting time between acceptance and loading, and often tighter cut-offs so that the goods spend as little time as possible on the ground. Worked Example 1 follows a perishable consignment through acceptance.

2. Temperature-sensitive pharmaceuticals, which add documentation and handling requirements proving and maintaining their storage conditions.

3. Live animals, valuables, and outsize items, each with prescribed packaging, documentation, and aircraft requirements.

4. Dangerous goods. Substances or articles that present hazards during transportation and are subject to applicable safety regulations. Lesson 3.3 introduced dangerous goods among the

cargo categories and their handling needs; air carriage applies the strictest version of those requirements. The dangerous goods classes run from Class 1, explosives, through Class 3, flammable liquids, Class 8, corrosives, and Class 9, miscellaneous articles including lithium batteries, and air carriage of such goods is governed by the air industry's dangerous goods regulations, which are based on the international technical instructions for the safe transport of dangerous goods by air and are updated annually.

Dangerous goods acceptance is a discipline of its own. The shipper must classify the goods, declare them on the dangerous goods declaration, use the required performance packaging, and apply the correct hazard labels. At acceptance, dangerous goods may be checked only by trained and qualified staff, who verify the declaration, packaging, marking, and labelling against the applicable checklists; some dangerous goods are restricted to freighter aircraft or excluded from air transport altogether. Worked Example 3 shows what happens when a shipper tries to skip all of this.

KEY CONCEPT

Unit load devices and build-up

Aircraft holds are curved, and their floors are fitted with restraint systems. Loose cartons loaded piece by piece would waste space, shift in flight, and delay departure, so air cargo travels consolidated in unit load devices. A unit load device, ULD, is an aircraft pallet or container used to consolidate and secure cargo, baggage, or mail for air transportation, and it is treated as equipment of the aircraft itself: built and certified for flight, and a damaged ULD is withdrawn from service rather than loaded.

Two broad families exist.

1. Aircraft containers, rigid boxes shaped to fit the fuselage contours of the hold in which they travel. The industry identifies common types by short codes; a standard lower-deck container, for example, is known by the code AKE.

2. Aircraft pallets, flat platforms on which cargo is built up, secured with nets or straps, and covered where required. A standard main-deck pallet is known by the code PMC.

Build-up is the terminal activity of assembling loose cargo into these devices. The ground handling agent sorts consignments by flight, builds each unit load to a plan respecting weight distribution, secures it, and records its contents so the load documentation agrees with the cargo loaded. Several shippers' consignments on one airline booking may share a unit load; the waybill structure described in the next concept records each consignment within it.

The shipper's stake is practical: packages must fit and survive being built into a unit load, which is why acceptance examines packaging for suitability for air carriage. The carton leaves the shipper and flies inside a certified unit load assembled by professionals at the terminal.

KEY CONCEPT

The air waybill, the cargo manifest, and the journey from acceptance to delivery

Lesson 6.2, Bill of Lading and Air Waybill, taught the air waybill's legal character: the AWB is a non-negotiable document that evidences the contract of carriage and acknowledges receipt of the cargo, and it is never a document of title. This lesson adds its operational life. The document travels the whole journey as data, and its structure follows the contracts. The master air waybill, MAWB, is issued by the airline or its authorised agent covering cargo accepted under the airline's contract of carriage; the house air waybill, HAWB, is issued by the freight forwarder to a shipper for an individual consignment. Where a forwarder consolidates several customers' goods under one airline booking, each shipper holds its HAWB while the airline's MAWB covers the consolidated consignment, and each waybill number identifies its consignment at every checkpoint from acceptance to delivery. Since 1 January 2019, under IATA Resolution 672, the electronic air waybill has been the default contract of carriage on enabled trade lanes, paper the exception: the content and character are unchanged, but the document exists as data moving ahead of the goods.

Alongside the waybills runs the cargo manifest, a document or electronic record listing cargo carried on a particular transport service. Each flight's manifest tells the receiving airport what is coming, and manifest data commonly reach the destination customs authority before the aircraft lands, supporting advance risk analysis and faster release.

The complete journey of a consignment assembles every concept in this lesson.

1. Booking confirmed for a named flight.
2. Delivery to the terminal before the acceptance cut-off.
3. Acceptance: document check and physical check, including any dangerous goods acceptance.
4. Security screening or acceptance under an approved shipper's status.
5. Build-up into unit load devices.
6. Loading and departure.
7. Arrival, breakdown, and notification of the consignee.
8. Import customs formalities and release.
9. Delivery to the consignee.

Compared with the ocean journey of Lesson 7.1, the pattern is identical and the timescale is compressed: weeks become hours, the vessel becomes the aircraft, the container becomes the ULD, and the documentation cut-off becomes the acceptance cut-off, closing shortly before the flight rather than days before the sailing. The next lesson, Containerisation: FCL, LCL, and Container Types, returns to the sea and examines the container itself.

WORKED EXAMPLE

A vegetable consignment accepted at Kotoka International Airport

A vegetable exporter based in Kumasi supplies fresh chillies and courgettes to a produce importer in Amsterdam. On Tuesday the importer requests a consignment of 30 cartons for Thursday's market. All figures and times in this example are illustrative for teaching purposes.

Step 1: Booking. The exporter's forwarder asks an airline for space on Thursday evening's departure from Kotoka International Airport, Accra, to Amsterdam, stating the consignment's details: 30 cartons, the weight and dimensions of a packed carton, the nature of the goods, and their perishable handling requirement. The airline confirms the space and gives the acceptance cut-off: Thursday afternoon, several hours before the late-evening departure, because the goods must be accepted, screened, and built up in good time.

Step 2: Preparation and documents. The cartons are packed on Thursday morning from the morning's harvest in vented, stackable cartons, marked with the exporter's and importer's details and package numbers, and labelled for perishable handling. The forwarder assembles the paperwork: air waybill data agreeing with the commercial invoice and packing list, and the phytosanitary certificate issued by the plant-protection authority, since plants and plant products require it.

Step 3: Delivery and document check. The refrigerated truck reaches the cargo terminal on Thursday afternoon, ahead of the cut-off. At acceptance, staff check the documents against the booking: parties, routing, 30 pieces, weight, description, and the phytosanitary certificate, all in order.

Step 4: Physical check. The cartons are counted, 30, and checked against the stated weight and dimensions. The packaging is examined and found sound and suitable for build-up into a unit load; the markings agree with the air waybill.

Step 5: Screening and build-up. The consignment's security status is established through screening, and the cartons are held in the cold room. The ground handling agent builds the consignment into an aircraft pallet with other cargo for the same flight, securing and recording the unit.

Step 6: Acceptance recorded, flight, and delivery. The consignment flies that evening. In Amsterdam the unit load is broken down overnight, the importer is notified, customs formalities follow, and the cartons reach the market within the window on Friday.

What does this example demonstrate? Acceptance is a sequence, not a stamp: documents, physical condition, screening, and build-up each had to pass before the cartons could fly, and every step sat inside the cut-off so that the cold chain never broke.

Key lesson: perishable air freight succeeds backwards, from the market window to the harvest, with the acceptance cut-off as the pivot the whole plan must respect.

WORKED EXAMPLE

Chargeable weight on a quotation desk in Shenzhen

A freight forwarder's quotation desk in Shenzhen handles two consignments for the same buyer in Accra, flying on the same service. All figures are illustrative for teaching purposes, and the IATA divisor of 6,000 applies.

Step 1: Consignment A, the dense carton. A manufacturer offers one carton of electric motor components, measuring 40 by 30 by 30 centimetres and weighing 18 kilograms. The volumetric weight is 40 multiplied by 30 multiplied by 30, which is 36,000 cubic centimetres, divided by 6,000: 6 kilograms. The actual weight of 18 kilograms is greater than the volumetric weight of 6 kilograms, so the chargeable weight is 18 kilograms. The carton is charged on what it weighs, because it exhausts the aircraft's payload, not its space.

Step 2: Consignment B, the bulky carton. A second manufacturer offers one carton of plastic householdware, measuring 60 by 50 by 40 centimetres and weighing 8 kilograms. The volumetric weight is 60 multiplied by 50 multiplied by 40, which is 120,000 cubic centimetres, divided by 6,000: 20 kilograms. This time the volumetric weight of 20 kilograms is greater than the actual weight of 8 kilograms, so the chargeable weight is 20 kilograms. The carton is charged on the space it occupies, because it exhausts the aircraft's volume long before its payload.

Step 3: The comparison. Consignment B weighs less than half of consignment A, yet its chargeable weight is higher: 20 chargeable kilograms against 18. Quoted by the same per-kilogram price for the flight, the lighter carton costs more to fly. The buyer's first reaction is disbelief; the explanation is the aircraft itself, which sells payload and volume together and charges whichever resource a consignment consumes more of.

Step 4: The desk's advice. The forwarder advises both manufacturers to state dimensions with every enquiry, and the bulky manufacturer to review its packaging: a carton redesigned to waste less space lowers its chargeable weight, because every cubic centimetre of air inside the packaging is billable volume.

What does this example demonstrate? Chargeable weight is the greater of actual and volumetric weight, and for light, bulky goods the dimensions, not the scales, decide the freight.

Key lesson: when you buy air freight, you are buying space as well as weight; measure your cartons before you price your shipment, because the airline certainly will.

WORKED EXAMPLE

Six drums refused: the acceptance check that catches a hidden dangerous good

A coatings manufacturer near Tema receives an urgent order from an industrial buyer overseas for six steel drums of a solvent-based coating material, 120 kilograms in total. A colleague books the consignment with a forwarder as ordinary general cargo, described simply as coating material. All details are illustrative for teaching purposes.

Step 1: Delivery and the document check. The drums reach the cargo terminal at Kotoka International Airport in time for the booked flight. The acceptance officer compares the air waybill data against the booking: parties, routing, six pieces, the weight, and the description, coating material. Nothing on the documents mentions dangerous goods, and no dangerous goods declaration is attached.

Step 2: The physical check raises the question. Counting the pieces, the officer finds six sealed steel drums. Drums of liquid chemical are exactly the presentation that acceptance training flags: the description says coating, and solvent-based coatings are typically flammable liquids, which fall under Class 3 of the dangerous goods classes. The packaging carries no hazard labels and no marks of the performance packaging required for dangerous goods.

Step 3: The refusal. The officer asks for the product's safety data sheet, the manufacturer's document that classifies the product's hazards and is the starting point for deciding whether goods are regulated for transport. It confirms a flammable liquid. Carrying the consignment as general cargo, undeclared, unlabelled, and in non-performance packaging, is not permissible, and the acceptance check refuses the consignment. The drums do not fly, whatever the departure schedule says.

Step 4: The correct path. The manufacturer starts again, properly. The product is classified from its safety data sheet as a Class 3 dangerous good. A forwarder with trained dangerous goods staff is engaged, the airline is told at booking that dangerous goods are offered, the drums are packed in the required performance packaging, marked with the proper shipping name and UN number, and labelled with the Class 3 hazard label, and the shipper completes the dangerous goods declaration. At acceptance, staff qualified for dangerous goods check the declaration, packaging, marking, and labelling against the checklist, and the consignment is accepted to travel as what it is.

What does this example demonstrate? Acceptance exists to catch exactly this situation. The refusal was not the system failing; it was the system working, and the checks, documentary, physical, and specialist, combined to stop an undeclared hazard from entering an aircraft's hold.

Key lesson: the shipper decides the description, so the shipper owns the danger; classify products honestly before offering them for air carriage, and route regulated goods through trained hands.

IMPORTANT NOTE

Quote dimensions, not just weight

The commonest expensive surprise in air freight is the shipper who prices a consignment on weight alone. The forwarder's quotation follows the chargeable weight, the greater of the actual and the volumetric weight, so a consignment of light, bulky goods can be quoted, booked, and then re-rated at the terminal when it is measured, leaving the shipper with a freight charge noticeably above the estimate the weight suggested.

The habit that prevents this costs nothing: state every package's length, width, and height, as well as its weight, in every enquiry and booking. The dimensions let the forwarder compute the volumetric weight, apply the applicable divisor, and quote the true chargeable weight before anyone commits. Irregularly shaped items occupy the space they measure at their greatest points, and palletised goods inherit the pallet's dimensions, so the outer envelope of the package is what the tape measure records.

Two corollaries are worth carrying with the habit.

1. Light goods are not cheap goods. In air freight the question is never only how heavy but how big. A consignment of plastic products can cost more to fly than a heavier consignment of steel parts, as this lesson's worked example showed.

2. Packaging design is freight-cost design. Air inside a carton is billable space. Reducing a bulky carton's dimensions, or reconfiguring packing so more product travels in less volume, lowers the chargeable weight directly.

When a quotation and an expectation diverge, ask which of the two weights drove the charge. The answer is almost always the volumetric one, and the tape measure explains what the scales never could.

IMPORTANT NOTE

The paperwork closes before the aircraft does

Passengers learn airport rhythms that do not transfer to cargo. A traveller can check in shortly before departure, and beginners reasonably imagine cargo boards the same way: deliver the goods to the airport an hour or two before the flight and all is well. The reality is the opposite, and it follows from the structure of this lesson. Cargo travels on passenger and freighter flights alike, but the paperwork and the operational preparation close first.

Consider what must happen between a consignment's arrival at the terminal and departure: the document check, the physical check, security screening, storage and, for perishables, cold storage, build-up of unit load devices, and loading to the aircraft's plan. Each step takes time, each is performed by different hands, and departure time is fixed. The acceptance cut-off is therefore set well before departure so the sequence fits inside it, and a consignment delivered after the cut-off has missed the flight even though the aircraft has not yet landed.

Missing a cut-off is not a fine and a smile at the desk. The consignment comes off the flight, and the forwarder must rebook it on a later departure, subject to space being available; on a busy lane the wait can stretch beyond the next scheduled service. For perishable goods the missed flight is not merely a delay but a deterioration risk, which is why perishable cut-offs are often the tightest of all.

The planning rule follows: work backwards from the cut-off, not from the departure time. Ask the forwarder for the acceptance deadline at booking, and treat that deadline, not the departure, as the moment the consignment must be ready, delivered, and documented.

IMPORTANT NOTE

Never sign a cargo description you cannot stand over

The shipper's declaration is the foundation the entire acceptance process stands on. Every check at the terminal compares the goods against what the shipper stated: the description on the transport document, the weights, the dimensions, and, critically, the assertion, explicit or implied, that the consignment is what it claims to be and is not a regulated dangerous good travelling undeclared. When a shipper signs a description, the shipper, not the forwarder and not the airline, is the party that put those words into the documents.

The gravest form of failure is the one Worked Example 3 dramatised: dangerous goods offered as general cargo. An undeclared flammable liquid in an aircraft hold is a threat to the aircraft and everyone on it, and misdeclared dangerous goods are treated with corresponding seriousness wherever air cargo is handled. The honest, safe, and in the end quicker path is always classification first: consult the product's safety data sheet, and if the product is regulated, declare it, pack it, and label it as the dangerous goods requirements demand, through people trained to handle it.

The same integrity applies to ordinary descriptions. Guessed weights, estimated piece counts, or padded descriptions are discrepancies waiting to surface at acceptance, where the physical consignment is compared with the document, and afterwards in customs processing at the destination, where the declaration must match the goods. Lesson 3.3, Cargo Types, Packaging, and Handling, explained why cargo type drives handling; lesson 6.4, Document Verification and Common Discrepancies, showed what document errors cost once they are found.

The professional habit is short enough to memorise: weigh, measure, classify, describe, and only then sign. A description you cannot stand over is not paperwork; it is a liability travelling with the aircraft.

Practice questions (Lesson Checks)

Practice Question 1

An exporter in Accra asks a forwarder to quote for flying two cartons to the same buyer on the same flight. Carton A contains machine parts and weighs 18 kg, measuring 40 x 30 x 30 cm. Carton B contains plastic householdware and weighs 8 kg, measuring 60 x 50 x 40 cm. The volumetric divisor is 6,000.

Which statement best explains which carton attracts the higher air freight charge, and why?

1. Carton A attracts the higher charge, because air freight is always calculated on the actual gross weight shown on the air waybill, and 18 kg is more than double 8 kg.

2. Carton B attracts the higher charge, because its volumetric weight is 20 kg (120,000 divided by 6,000), which exceeds its actual weight of 8 kg, so its chargeable weight of 20 kg exceeds Carton A's chargeable weight of 18 kg. ✓

3. Both cartons attract the same charge, because the airline adds the two actual weights together and prices the two cartons as one 26 kg consignment.

4. Carton B attracts the higher charge, because plastic goods are classified as special cargo and carry a surcharge on international flights.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Air freight is charged on chargeable weight, the greater of the actual gross weight and the volumetric weight, because an aircraft sells payload and volume together and each consignment pays for whichever resource it consumes more of. Carton A's volumetric weight is 40 x 30 x 30 divided by 6,000, which is 6 kg, so its actual weight of 18 kg governs and its chargeable weight is 18 kg. Carton B's volumetric weight is 60 x 50 x 40 divided by 6,000, which is 20 kg, so its volume governs and its chargeable weight is 20 kg. The lighter carton is charged more, which is exactly the surprise this lesson warns beginners about.

Answer 1 fails because air freight has never been charged on actual weight alone; volumetric weight exists precisely to price space. Answer 3 fails because each consignment's chargeable weight is assessed on the comparison between its own actual and volumetric weights, not by merging shipments into one weight. Answer 4 fails because plastic householdware is ordinary general cargo; the charge follows the dimensions, not the material.

Takeaway: in air freight the tape measure can outvote the scales; measure and weigh every package before pricing a shipment.

Practice Question 2

A trading company delivers 15 cartons of general cargo to the airline's cargo terminal at 17:00 for an aircraft departing at 23:00 the same evening. The company's manager believes six hours is ample, but the terminal explains that the consignment cannot travel on that flight.

Which statement best explains the position?

1. The consignment cannot travel because general cargo is only carried on dedicated freighter aircraft, and the 23:00 departure is a passenger service.

2. The consignment cannot travel because acceptance for the flight closed earlier: cargo must be delivered, document-checked, screened, and built into unit load devices before the aircraft is loaded, so the operational clock closes well before departure. ✓

3. The consignment cannot travel because airlines accept only consolidated loads containing many shippers' goods, and 15 cartons from one shipper is too small a booking.

4. The consignment will still travel if the company pays a late-delivery charge at the desk, since the aircraft's hold space is confirmed at booking.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The acceptance cut-off closes well before departure because a whole sequence must fit inside the time available: the document check, the physical check, security screening or verification of approved security status, storage, build-up of unit load devices, and loading to the aircraft's plan. The aircraft departs on schedule whether or not a late consignment has completed that sequence, so the paperwork and the load plan close first, and cargo can miss a flight that has not yet departed. The remedy is rebooking on a later departure, subject to space.

Answer 1 fails because cargo regularly travels in the holds of passenger aircraft; the flight type is not the obstacle. Answer 3 fails because small consignments fly routinely, often consolidated with others, and size is not the obstacle. Answer 4 fails because the cut-off is an operational deadline, not a payment threshold: no fee converts a consignment that missed acceptance, screening, and build-up into cargo that is safely loaded.

Takeaway: plan backwards from the acceptance cut-off, not from the departure time, and ask for the cut-off at the moment of booking.

Practice Question 3

A consignment is presented for carriage by air and passes through the checks performed before an airline will accept it. A colleague who has not studied air freight makes four statements about the process.

Which statement is NOT correct?

1. Acceptance includes checking that the number of pieces, the weight, and the dimensions agree with the air waybill and the booking.

2. Acceptance includes examining the packaging, marking, and labels for suitability for air carriage, including hazard labels for any dangerous goods.

3. Acceptance is finished as soon as the forwarder collects the goods from the shipper's premises, and the airline or its ground handling agent performs no checks of its own at the terminal. ✓

4. Dangerous goods may be accepted only by trained and qualified staff who check the declaration, packaging, marking, and labelling against the applicable requirements.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

Acceptance is performed by the airline or its ground handling agent at the cargo terminal, and it is the gatekeeping function on which the safety, security, and document integrity of the flight depend. Collection from the shipper is only the delivery of the consignment to the start of the process; the document check, the physical check of pieces, weights, dimensions, packaging, marking, and labelling, security screening, and any dangerous goods acceptance all happen at or through the terminal afterwards. A forwarder's collection involves none of those checks.

Answer 1 is correct as a description of acceptance, because the physical consignment is compared against the documents the booking and chargeable weight were built on. Answer 2 is correct, because packaging suitability and correct marking and labelling are standard elements of the physical check. Answer 4 is correct, because dangerous goods acceptance is a specialist task reserved for trained, qualified staff, as this lesson taught and as Worked Example 3 showed.

Takeaway: acceptance is the airline's gate, checked at the terminal against documents and physical reality; nothing a forwarder does before delivery replaces it.

Fact-check appendix (audit notes — not for entry)

Definitions of Air Cargo, Air Freight, Air Cargo Terminal, Ground Handling Agent (GHA), Cargo Acceptance, Air Waybill (AWB), Master Air Waybill (MAWB), House Air Waybill (HAWB), Air Cargo Screening, Unit Load Device (ULD), Chargeable Weight, Special Cargo, Perishable Cargo, Dangerous Goods, and Cargo Manifest follow the lesson's key terms — source: curriculum/lesson_briefs/module-07/lesson-7.2.md key terms.

Chargeable weight is the higher of gross weight and volumetric weight; IATA volumetric formula: volumetric weight (kg) = L x W x H in centimetres / 6,000 (6,000 cubic cm = 1 kg); some carriers/contracts use 5,000 or 7,000 as contractual divisors, 6,000 being the IATA standard — source: trade-transport-standards.md section 8 (air cargo).

ULDs are aircraft containers and pallets that consolidate cargo for loading, certified under the IATA ULD Regulations and treated as aircraft equipment; common examples AKE (LD3-type lower-deck container) and PMC (96 x 125 inch pallet); lesson keeps the treatment light — source: trade-transport-standards.md section 8 (ULD), per the lesson guidance to keep ULD detail light.

e-AWB: under IATA Resolution 672 (Multilateral e-AWB Agreement) the electronic AWB has been the default contract of carriage since 1 January 2019 on enabled trade lanes, paper the exception — source: trade-transport-standards.md sections 4 and 8.

MAWB issued by the airline or its authorised agent; HAWB issued by the freight forwarder to the shipper for an individual consignment — source: trade-transport-standards.md section 4 and the lesson brief key terms.

The AWB is always non-negotiable (receipt, evidence of contract, freight bill, customs document, never a document of title); its full legal character was taught in lesson 6.2 and only referenced here — source: trade-transport-standards.md section 4.

Cargo acceptance basics: confirmed booking/AWB, security screening/status under national regulated-agent regimes, acceptance checks (marking, packaging, ULD build-up), and DGR acceptance by trained, authorised staff against checklists — source: trade-transport-standards.md section 8 (cargo acceptance basics).

Known-consignor concept taught generically (security-approved shippers whose cargo enters the air side on the strength of their approved supply-chain controls, versus cargo screened before acceptance); no country programme names used — source: trade-transport-standards.md section 8 (security screening/status under national regulated-agent regimes), generalised per the lesson guidance; the term "known consignor" as a generic industry expression is general industry knowledge, flagged for fact-checker.

Dangerous goods classes 1-9 (including Class 3 flammable liquids, Class 8 corrosives, Class 9 miscellaneous including lithium batteries); UN numbers with proper shipping names; UN-performance packaging and hazard labels; shipper's modal dangerous goods declaration (Shipper's Declaration for Dangerous Goods under IATA DGR) stating UN number, proper shipping name, class, packing group, quantities and packing/label compliance — source: trade-transport-standards.md sections 5 and 9.

IATA DGR is the air modal rulebook, based on the ICAO Technical Instructions and updated annually — source: trade-transport-standards.md section 9.

SDS/MSDS is the 16-section GHS document from the manufacturer and the starting point for deciding if goods are regulated for transport, but the SDS itself is not the transport declaration — source: trade-transport-standards.md section 9.

Some dangerous goods are restricted to freighter aircraft or excluded from air transport altogether — general industry knowledge, stated qualitatively without specifics; flagged for fact-checker.

Phytosanitary certificates are required for plants and plant products and issued by the exporting country's plant-protection authority in IPPC model format; health/sanitary certificates relate to products of animal origin — source: trade-transport-standards.md section 5.

Manifest data reach the destination customs authority before arrival, supporting advance risk analysis and faster release (pre-arrival declaration concept) — source: trade-transport-standards.md section 12, taught qualitatively.

Booking flow (enquiry, booking, booking confirmation, preparation, delivery within cut-off) parallels the ocean booking flow — source: trade-transport-standards.md section 7 (booking flow), transposed to air as general industry practice; flagged for fact-checker.

Cut-off concept (acceptance closes well before departure; times vary by airline, airport, flight type and commodity; no universal figures stated; all times in examples labelled illustrative) — source: general industry practice, taught qualitatively per WRITING_SPEC.md rule 5.1; flagged for fact-checker.

Air cargo travels in the holds of passenger aircraft and on dedicated freighters; the air product is airport-to-airport with road legs completing the journey; road feeder services carry cargo between airports under air cargo documents — general industry knowledge, qualitative only; flagged for fact-checker.

Reasons shippers choose air (speed, security, shorter in-transit exposure of capital and insurance risk) and the air-suitable commodities (perishables, pharmaceuticals, electronics, spares, samples) — general business knowledge, qualitative; consistent with the lesson guidance.

Cargo passes export and import customs formalities; authorities apply commodity-specific controls — general business knowledge, qualitative.

ULD build-up by the ground handling agent with weight-distribution planning and unit-load records — general industry knowledge, qualitative; flagged for fact-checker.

Screening by scanning equipment and physical examination; unscreened/unknown-status cargo is not loaded until its status is established — general industry knowledge, qualitative; flagged for fact-checker.

Packaging must withstand unit-load handling and stacking; air packaging is designed light; markings must agree with the transport document; handling labels indicate orientation/fragility generically — general industry knowledge, qualitative; flagged for fact-checker.

Re-rating of cargo at the terminal when measured figures differ from declared figures; packaging redesign reduces chargeable volume — general industry practice, qualitative; flagged for fact-checker.

Ghanaian and international settings: Kumasi packhouse, Kotoka International Airport as Ghana's air cargo gateway, Tema, Accra; international settings Amsterdam and Shenzhen; goods: fresh vegetables, machine parts, plastic householdware, solvent-based coating material — settings and goods per WRITING_SPEC.md section 4 (Kotoka International Airport in the approved settings list); airline unnamed and no AWB number quoted, respecting the lesson guidance and DO NOT CLAIM items 1 and 5.

No duty rates, tax percentages, levies, fee amounts, statutory deadlines, free-time periods, freight rates, per-kilogram prices, or e-AWB adoption percentages appear anywhere in the lesson; all weights, dimensions, times, and commercial details in worked examples are illustrative for teaching purposes — source: WRITING_SPEC.md rule 5.1 and DO NOT CLAIM item 5 (respected).

Referenced lessons by name only, without re-teaching: lesson 2.2 Freight Forwarders, Carriers, Shipping Lines, and Agents; lesson 3.3 Cargo Types, Packaging, and Handling (cargo categories and dangerous goods introduction); lesson 5.1 Role and Responsibilities of Freight Forwarders; lesson 6.2 Bill of Lading and Air Waybill (AWB legal character, MAWB/HAWB, e-AWB); lesson 6.4 Document Verification and Common Discrepancies; lesson 7.1 Ocean Freight Operations and Vessel Schedules (ocean booking, cut-offs, container terminal comparison); next lesson 7.3 Containerisation: FCL, LCL, and Container Types — source: curriculum/lesson_briefs placements.

Lesson 7.3 — Containerisation: FCL, LCL, and Container Types

Summary

This lesson examines containerisation and its role in modern international cargo transportation. Learners will explore how standardised shipping containers facilitate cargo handling, storage, and multimodal transportation. The lesson explains the differences between full container load and less than container load shipments, introduces common container types and sizes, and examines the basic procedures involved in container selection, loading, securing, and movement.

Learning objectives

- Define containerisation and explain its importance in international logistics.
- Identify the advantages of using standardised shipping containers.
- Distinguish between full container load and less than container load shipments.
- Identify common container types and their appropriate cargo applications.
- Explain the differences between standard, high-cube, refrigerated, open-top, and flat-rack containers.
- Describe the basic procedures involved in container loading and cargo securing.
- Explain the purpose of container identification numbers and security seals.
- Identify factors influencing container selection for different shipments.
- Describe the role of container freight stations in consolidated cargo operations.
- Recognise common operational and safety considerations associated with containerised transportation.

Key terms

Containerisation — The use of standardised freight containers to transport goods efficiently across different transportation modes.

Shipping Container — A reusable transport unit designed for the secure and efficient movement of cargo.

Full Container Load (FCL) — A container shipment generally arranged for the use of one shipper or consignee rather than being consolidated with unrelated shipments.

Less than Container Load (LCL) — A shipment occupying part of a container and typically consolidated with cargo belonging to other shippers.

Twenty-Foot Equivalent Unit (TEU) — A standard measurement of container capacity equivalent to one nominal 20-foot shipping container.

Forty-Foot Equivalent Unit (FEU) — A container capacity measurement equivalent to one nominal 40-foot shipping container, generally equal to two TEUs.

Dry Container — An enclosed container designed for transporting general cargo that does not require temperature control.

High-Cube Container — A container with greater internal height than a corresponding standard-height container.

Refrigerated Container (Reefer) — A temperature-controlled container used to transport goods requiring specified temperature conditions.

Open-Top Container — A container with a removable or flexible roof covering, suitable for certain oversized or top-loaded cargo.

Flat-Rack Container — A container platform with end structures, designed for transporting certain oversized or unusually shaped cargo.

Container Seal — A security device used to indicate whether a container door or other secured access point has been opened or tampered with.

Container Number — A standardised identification code assigned to a freight container.

Container Freight Station (CFS) — A facility used for receiving, consolidating, deconsolidating, and handling containerised cargo.

Cargo Securing — The process of restraining cargo to prevent unsafe movement or damage during transportation.

Verified Gross Mass (VGM) — The verified total gross mass of a packed container determined in accordance with applicable SOLAS requirements.

Teaching content

KEY CONCEPT

What Containerisation Changed

Containerisation is the use of standardised freight containers to transport goods efficiently across different transportation modes. A shipping container is a reusable transport unit, built to a common international specification, designed for the secure and efficient movement of cargo.

To appreciate what changed, picture ocean cargo before containers. A consignment of mixed goods travelled as break-bulk: individual crates, drums, cartons and bales lifted aboard one by one, stowed by hand and hook deep in a vessel's holds, then lifted out again at the destination port and handled once more at each warehouse. Every piece was touched many times, and every touch was a chance for damage, loss, delay and theft.

Containerisation replaced the handling of cargo with the handling of a box. Goods are packed once, at the origin, into a steel container; from that moment the container itself is the unit that everyone lifts. A crane takes it from truck to ship, the ship carries it across the ocean, another crane sets it on a chassis or a rail wagon at the destination port, and the cargo inside is never touched until the box is opened at its destination. This is the intermodal and multimodal movement introduced in Lesson 3.2, Modes of Transport: Sea, Air, Road, Rail, and Multimodal, and the container is the unit that makes it work smoothly.

The consequences reach every part of international logistics.

1. Less handling damage. Cargo packed once into a sealed box is protected from repeated lifting, stacking and weather exposure.

2. Faster port turns. A crane moves one twenty-tonne box in a single lift instead of dozens of separate pieces, so vessels spend hours at the berth instead of days.

3. A sealable customs unit. The container is a sealable customs unit: it is sealed after stuffing, the seal number is recorded, and customs can track and inspect a sealed, identified box rather than loose cargo. Sealed containers can even cross borders in transit without their contents being handled.

Containerisation is therefore not merely a type of equipment. It is the standard that reshaped world trade by making the sealed box, not the individual package, the unit of carriage.

KEY CONCEPT

Standardisation: ISO Dimensions and the Language of TEU and FEU

The power of the container lies in its standardisation. Freight containers are built to the international standards ISO 668, which sets their classification, dimensions and ratings, and ISO 1496, which sets their specification and testing. Because every container shares the same corner fittings, the same lifting spreaders grip boxes in Accra, Antwerp and Singapore alike; the same trucks, rail wagons and vessel cell guides accept them everywhere. One box moves from a factory floor to a truck to a ship to a train without any modification. That is the whole point of the standard.

Standardisation also created a common language for measuring volume, and learners must speak it.

1. TEU. The twenty-foot equivalent unit is the standard measurement of container capacity, equivalent to one nominal 20-foot shipping container. Vessel sizes, port throughputs and fleet capacities are quoted in TEUs: a container ship described as 15,000 TEU could, in principle, carry that many nominal twenty-foot boxes.

2. FEU. The forty-foot equivalent unit is a container capacity measurement equivalent to one nominal 40-foot shipping container, generally equal to two TEUs. A single forty-foot box is one FEU, or two TEUs; two twenty-foot boxes are also two TEUs. A forty-foot high-cube container, despite its extra height, still counts as one FEU, because the unit measures the nominal length, not the volume.

In practice, forty-foot containers dominate many trade lanes, so a booking for two FEUs usually means two forty-foot boxes rather than four twenty-foot boxes. When Lesson 7.4, Shipping Charges, Freight Rates, and Surcharges, discusses how freight is priced, learners will see that ocean rates for containerised cargo are commonly quoted per container, and the TEU and FEU are the units in which ships, ports and the industry count their business.

KEY CONCEPT

The Dry Container: Internal Space and the Weight-Volume Trade-Off

The dry container is an enclosed container designed for transporting general cargo that does not require temperature control. It is the steel box everyone pictures: steel doors at one end, a wooden or bamboo floor, and no machinery of any kind. Dry containers carry the overwhelming

majority of the world's containerised cargo, from cartons of consumer goods to bagged rice to spare machinery parts.

The two common lengths have internal dimensions that every practitioner should carry in the head, at least approximately. The figures below are typical values for standard boxes; an individual container's data plate is always the authority for that particular box.

1. The 20-foot dry container. Internally approximately 5.898 metres long, 2.352 metres wide and 2.393 metres high, giving about 33 cubic metres of space. Its maximum gross mass, the rated total of container plus cargo, is 30,480 kilograms. With an empty container weight, called the tare, of approximately 2,230 kilograms, the payload, the maximum permissible cargo weight, is approximately 28,250 kilograms.

2. The 40-foot dry container. Internally approximately 12.032 metres long, 2.352 metres wide and 2.393 metres high, giving about 67.5 cubic metres, double the twenty-foot's volume. But the permitted weight does not double: its max gross is commonly rated at the same 30,480 kilograms, and with a tare of roughly 3,750 to 3,800 kilograms its payload is approximately 26,700 kilograms, slightly less than a twenty-foot's.

This asymmetry is the single most useful rule in container selection: a forty-foot box doubles the volume but not the permitted weight. Dense, heavy cargo, such as machinery parts, ingots or bagged cement, fills a twenty-foot's weight allowance long before it fills its volume, so heavy cargo travels in twenty-footers. Light, bulky cargo, such as plastic household goods or empty packaging, exhausts a forty-foot's volume long before its weight allowance, so light bulky cargo travels in forty-foot and high-cube boxes. Module 9 returns to what happens when boxes sit full of cargo but late or uncollected; for now, note only that choosing the wrong pairing of cargo and box wastes money on either side of the trade-off.

KEY CONCEPT

High-Cube and Refrigerated Containers

Two variations of the standard box cover the commonest special needs: cargo that is bulky but light, and cargo that must be kept cold.

1. The high-cube container. A high-cube container has greater internal height than a corresponding standard-height container. The standard forty-foot box stands 8 feet 6 inches tall externally; the forty-foot high-cube stands 9 feet 6 inches, about 2.896 metres, externally, with an internal height of approximately 2.698 metres. The footprint is identical to the standard forty-foot, but the extra height raises the volume to roughly 75 to 76 cubic metres. High cubes therefore suit light, bulky cargo that can be stacked high: furniture, insulation materials, plastic articles, garments hung high on racks. Because the footprint and weight rating are essentially the same as the standard box, the high-cube gains volume without gaining useful weight capacity.

2. The refrigerated container, or reefer. A refrigerated container is a temperature-controlled container used to transport goods requiring specified temperature conditions. It is an insulated box with an integral refrigeration unit built into its front wall. Reefers carry perishable foods, such as fruit, vegetables, fish and meat, and temperature-sensitive pharmaceuticals. The required temperature is set on the unit's controller before loading and monitored throughout the journey; modern units log the temperature continuously so that the whole journey's cold chain can be evidenced afterwards. The reefer does not make its own cold in a vacuum: it must be supplied with electrical power, so it is plugged into power at the packing point and at the terminal, and on board the vessel it runs on the ship's electrical supply. Reefer containers are a minority of the world fleet, roughly six per cent, and they receive closer attention everywhere they pass, because a failed reefer is a failed cargo.

Because insulation and machinery occupy space, a reefer offers somewhat less internal volume than a dry box of the same nominal length, a point that matters when planning how many cartons will fit.

KEY CONCEPT

Open-Top, Flat-Rack, and ISO Tank Containers

When cargo will not go through a standard box's doors, or will not fit its shape, three specialised container types provide the answer.

1. Open-top container. An open-top container has a removable or flexible roof covering, in practice a strong tarpaulin stretched over a bows framework, suitable for certain oversized or top-loaded cargo. Cargo that is too tall for a dry container's door opening, or too awkward to load through doors at all, is lowered in from above by crane, and the tarpaulin is then secured over the top. Typical open-top cargo includes machinery taller than the door aperture, sheet materials, and long items that must lie flat along the full length.

2. Flat-rack container. A flat-rack container is a platform with end structures, the fixed or collapsible end walls called headers, designed for transporting certain oversized or unusually shaped cargo. It has no sides and no roof. Flat racks carry items that are wider or longer than a box, or simply not rectangular: construction machinery, vehicles, industrial vessels, transformers, yachts. Such cargo is often described as out-of-gauge, meaning it projects beyond the dimensions a standard box would allow. Cargo lashed to a flat rack is exposed, so weather protection and securing are planned as part of the loading.

3. ISO tank container. An ISO tank is a tank mounted within a standard twenty-foot frame, used for liquids, gases and powders, and very often for dangerous goods. Tank containers are a small share of the fleet, under one per cent, but they are the standard answer for bulk liquids. A liquid product can be drummed and palletised into a box, or shipped in far larger quantity per movement in a tank; a bulk quantity of a natural oil such as the shea butter met in Lesson 3.3, Cargo Types, Packaging, and Handling, could travel either way, and the tank suits large, regular,

liquid volumes. Tank containers are cleaned between trips and certified for the product they carry.

KEY CONCEPT

Full Container Load: The Shipper's Own Sealed Box

A full container load, FCL, is a container shipment generally arranged for the use of one shipper or consignee rather than being consolidated with unrelated shipments. In an FCL movement, the customer's cargo alone travels inside the box.

The operational sequence is what defines the term. The ocean carrier tenders an empty container to the shipper or its agent. The shipper stuffs it, at its own premises or at a yard, with its own cargo only; the cargo is arranged, secured and counted; the doors are closed; and a seal is fixed through the door handles. The number of that seal is recorded on the shipping documents. From that moment the carrier receives and carries a sealed box and delivers it, on the export voyage, to the destination port or terminal still sealed. Nobody in the chain has opened the box between stuffing and arrival. Lesson 7.1, Ocean Freight Operations and Vessel Schedules, described the booking and cut-off clocks this box must meet.

Two clarifications prevent common misunderstandings.

1. FCL describes exclusivity, not fullness. A full container load does not have to be a full container. If a shipper books a twenty-foot box and fills only two-thirds of it, the shipment is still FCL, because no other party's cargo travels in that box. The term describes the commercial arrangement, exclusive use, not the fill level.

2. The container belongs to the carrier. The box is the shipping line's equipment, lent to the shipper for stuffing and returned empty after deconsolidation at destination. The shipper is responsible for it during its time in the customer's hands, a responsibility Module 9 examines through the charges that follow late returns.

FCL is the arrangement of choice where the cargo volume justifies a box, where the shipper wants control over how cargo is packed, and where minimal handling is worth arranging.

KEY CONCEPT

Less than Container Load and the Container Freight Station

A less than container load, LCL, shipment is a shipment occupying part of a container and typically consolidated with cargo belonging to other shippers. The consolidation work itself, grouping several customers' consignments into one box, was taught in Lesson 5.3, Shipping Instructions and Cargo Consolidation; this lesson looks at where it happens and what it means operationally for the cargo.

The facility at the centre of LCL is the container freight station, CFS, defined as a facility used for receiving, consolidating, deconsolidating, and handling containerised cargo. The CFS is the

warehouse where loose, palletised or crated cargo is received from many shippers, built up into containers, and, at the destination end, where the same containers are opened and the consignments separated again for collection.

An LCL consignment therefore passes through this sequence.

1. Delivery to the origin CFS. The shipper delivers the cargo, properly packaged and marked, to the CFS by the consolidator's cut-off, which falls before the container must reach the port terminal.

2. Stuffing with others. CFS staff load the consignment into the container together with other shippers' goods, planning the load so that cargo for different consignees can be separated and recovered at destination.

3. Sealed carriage. The consolidated container is sealed and moves as one unit through the port and across the ocean.

4. Deconsolidation at destination. At the destination CFS, the box is opened and the individual consignments are sorted out, checked and held for collection by each consignee or its agent.

The consequences for the cargo owner follow directly from this sequence. The cargo is handled at least twice more than FCL cargo, once at stuffing and once at deconsolidation. It shares the box with strangers' goods. It spends time at the CFS before sailing and again after arrival before it is available. And its packaging must survive its neighbours: a consignment packed in flimsy cartons travels beside other people's crated machinery.

KEY CONCEPT

The FCL or LCL Decision

Choosing between FCL and LCL is a decision, not a reflex. Neither option is universally better; the correct answer depends on the shipment. Five factors drive the choice.

1. Volume against a full box. Estimate the consignment's volume and weight and compare it with the space a twenty-foot or forty-foot box offers, roughly 33 and 67.5 cubic metres. A consignment that fills most of a twenty-foot is a natural FCL. A consignment of one or two pallets is a natural LCL. The middle ground, where the cargo fills a useful fraction of a box, is where judgement is needed.

2. Cost per unit of cargo. Lesson 5.3 introduced the cost logic qualitatively: LCL cargo pays a share of the box, commonly assessed on volume or on weight, plus CFS handling charges at both ends, while FCL pays for the whole box whether it is full or not. At low volumes the LCL share usually costs less in total; as volume grows, the accumulating shares eventually exceed the price of an exclusive box, and the crossing point is reached well before the box is actually full. No figures are quoted here because rates vary continuously; the lesson to hold is the shape of the comparison.

3. Handling risk. Every extra handling cycle is a chance of damage. FCL cargo is stuffed once and untouched until destination. LCL cargo is handled at stuffing, at deconsolidation, and possibly during any transfer, so fragile or retail-ready packaging suffers more in LCL.

4. Timing. FCL lets the shipper stuff at its own convenience, within the carrier's cargo cut-off. LCL requires delivery to the CFS by an earlier cut-off, surrendering some control of the schedule to the consolidator.

5. Cargo sensitivity. Goods that taint easily, attract theft, or must not travel beside certain neighbours are better protected in an exclusive, shipper-sealed box.

Weighing these five factors against the actual shipment is the professional habit this lesson wants to build.

KEY CONCEPT

Choosing the Right Container Type for the Cargo

The FCL or LCL decision settles who uses the box. A second, separate decision settles which box: container types exist because cargoes differ. Selection is driven by the cargo's own characteristics.

1. Temperature requirement. Any cargo needing specified temperature conditions, from frozen fish to chilled fruit to certain medicines, requires a reefer, with the set point agreed in advance. General cargo with no temperature need takes a dry container.

2. Volume and weight together. Dense cargo exhausts a twenty-foot's payload before its volume; light bulky cargo exhausts a forty-foot's or high-cube's volume long before any weight limit. Compare cubic metres and kilograms against the figures taught earlier for each box before choosing.

3. Dimensions and shape. Cargo taller than a door aperture or loadable only from above points to an open-top; cargo that is overheight, overwidth, overlength or irregular, out-of-gauge cargo, points to a flat rack; neither a high-cube nor any other box substitutes for the right specialised equipment.

4. Physical state. Liquids, gases and powders in quantity travel in ISO tank containers rather than in drums, wherever the volume justifies it.

5. Compatibility and regulation. Cargo that taints, or that must be segregated from other goods, including regulated dangerous goods with their own equipment and segregation requirements, constrains both the container type and what may travel inside it.

The selection questions compound: a light, bulky, temperature-sensitive pharmaceutical consignment needs a high-cube reefer, if such equipment is offered on the lane, or a standard reefer with a carefully planned load. The professional sequence is therefore: characterise the

cargo first, then choose the box, and only then take up the FCL or LCL question, because the cargo's needs bound both decisions.

KEY CONCEPT

Stuffing, Weight Discipline, and Cargo Securing

A packed container is a sealed promise, and the promise is only as good as the stuffing. Cargo securing is the process of restraining cargo to prevent unsafe movement or damage during transportation. It begins before the first carton enters the box and is the shipper's responsibility in FCL movements.

The rules of competent stuffing are few and strict.

1. Distribute the weight evenly. The cargo's weight should sit spread across the container floor, roughly centred over its length, not concentrated at one end or one side. An unevenly loaded box is dangerous: it lifts awkwardly, travels badly on a chassis, and can sit outside the limits a vessel's lashing plan assumes. Heavy items belong low in the box and over the floor, with lighter cargo above.

2. Respect the payload. The payload taught earlier is a rated limit, not a suggestion; a note at the end of this lesson returns to it in detail.

3. Secure the load inside the box. During a voyage a ship rolls and pitches, and a truck brakes and corners; loose cargo becomes a projectile. Cargo is braced with timber or inflatable bags in the gaps, restrained with lashing straps to the built-in lashing points, and packed tight where the cargo allows. Dunnage, the packing material placed to fill voids, protect surfaces and restrain movement, is the stuffing team's everyday tool.

4. Do not mix incompatible cargo. Strong-smelling goods taint foodstuffs and packaging; damp goods spread moisture to neighbours; dangerous goods may be segregated from each other and from other cargo by regulation. If a box must carry mixed cargo, compatibility is checked before the doors close, because afterwards nothing can be changed for weeks.

5. Declare the verified weight. Lesson 7.1 taught the verified gross mass rule: the shipper must determine and communicate the verified gross mass of the packed container before it is loaded, and a container without VGM does not sail. Accurate VGM begins with knowing what has been stuffed.

KEY CONCEPT

Container Numbers and Seals

Two pieces of identification turn a steel box into a controllable, documentable unit: the container number and the seal.

The container number is a standardised identification code assigned to a freight container. It is painted on the doors and on the sides and roof of the box, and it is unique to that physical container for its working life. Its purposes are practical and pervasive.

1. Identification across the chain. Booking, shipping instruction, bill of lading, customs declaration, terminal list and truck manifest all cite the same container number, so every participant, on every continent, knows precisely which box is meant.

2. Tracking and equipment control. The number is how a carrier traces its equipment and how a forwarder follows a shipment through ports and terminals; the unit moving past a gate camera is a number before it is a box.

3. Verification at stuffing. The professional checks that the number of the box actually stuffed matches the number on the booking and the documents. Stuffing the right cargo into the wrong box, or recording the wrong number, creates discrepancies discovered at the worst possible moment, at a border.

The container seal is a security device used to indicate whether a container door or other secured access point has been opened or tampered with. A seal is a numbered lock fixed through the closed door handles after stuffing, and its number is recorded on the shipping documents. On arrival, the interested parties compare the seal on the doors with the number on the documents: an intact seal bearing the recorded number is strong evidence that the doors were not opened during the journey, while a broken, missing or mismatched seal is recorded before the box is opened, noted on the receiving documents and notified to the carrier and, where relevant, customs. A seal is evidence of tampering, not prevention of it; it tells you that something happened, after the fact. Customs also uses seals for its own purposes, applying its own seal when cargo moves under customs control, for example in transit.

WORKED EXAMPLE

Choosing Between LCL and a Twenty-Foot Container in Accra

A producer of packaged shea butter cosmetics in Accra has received an order from a buyer in Antwerp. The finished order stands ready as nine pallets of cartons, approximately 4,800 kilograms in total weight and about 19 cubic metres in volume. The exporter must decide between LCL and a twenty-foot container. All figures in this example are illustrative for teaching purposes.

Step 1: Establish the facts. The exporter writes down the consignment's reality: nine pallets, 4,800 kilograms, about 19 cubic metres, retail-ready cartons that must reach a shop shelf looking untouched, and a production finish date that leaves a few days before the vessel the buyer expects.

Step 2: Compare with a full box. A twenty-foot container offers about 33 cubic metres and a payload of roughly 28,000 kilograms. Nine pallets fill well over half its floor area with room to stack lightweight cartons above, yet use less than a fifth of the weight allowance. The

consignment is too large to shrug off as a small parcel and too light to worry the box; volume, not weight, is the dimension at play.

Step 3: Apply the decision factors. On cost, the forwarder explains the shape of the comparison: LCL would bill the 19 cubic metres as a share of a box plus CFS handling charges at both ends, and at this size the accumulating share comes close to the whole-box price quoted for the twenty-foot. On handling, LCL means two extra handling cycles, at stuffing and deconsolidation, with strangers' cargo beside retail cartons. On timing, LCL requires delivery to the consolidator's CFS by an earlier cut-off, while a twenty-foot could be stuffed at the exporter's own premises the day production finishes. On sensitivity, the cartons are vulnerable to taint and crushing.

Step 4: Decide and execute. The exporter books the twenty-foot container. The box is delivered to the factory, stuffed the day production ends, pallets distributed across the floor, gaps braced with dunnage, heavy cartons low. The doors are closed, a seal is fixed, and the container and seal numbers are recorded on the documents. The verified gross mass is communicated to the carrier, and the box reaches the terminal before the cargo cut-off.

What does this example demonstrate?

The decision was not LCL or FCL by habit. It was the five factors, volume against a full box, cost per unit, handling risk, timing and cargo sensitivity, applied to one shipment.

Key lesson: in the middle volumes, the qualitative factors, especially handling risk and control of timing, frequently decide the choice; the exporter paid for exclusivity and bought a safer, better-scheduled shipment.

WORKED EXAMPLE

Stuffing a Reefer for Frozen Tuna: Mumbai to Rotterdam

A seafood processor in Mumbai has a contract to supply frozen tuna loins to a distributor in Rotterdam. The consignment will travel in a forty-foot reefer container, trucked to the port and shipped to Rotterdam. This example follows the stuffing, and then considers what would have happened if it had been done carelessly.

Step 1: Agree the carriage conditions. The shipping instructions specify the set point: minus 18 degrees Celsius, a widely used carriage temperature for frozen cargo. The carrier is told the set point when the reefer is booked, because the box must be allocated and prepared accordingly.

Step 2: Pre-cool the container. Before loading, the empty reefer is switched on and run until its interior has reached the set point. A reefer is built to hold a temperature, not to pull a warm load down to it. Loading warm cargo into a warm box hands the refrigeration unit an impossible job.

Step 3: Load from the cold store, fast. The tuna loins arrive at the packing point already at carriage temperature, straight from cold storage. The loading team works quickly to keep the

doors open for the shortest time, and places the cartons so that air can circulate: the chilled air must flow around and along the cargo, so the floor air channels are left clear, cartons are not stacked above the marked load line, and no wrapping or loose sheet is allowed to block the air path from the refrigeration unit. Frozen cargo is carried with the airflow moving around the cargo, not forced through it.

Step 4: Close, seal, record and power. The set point and the cargo temperature are checked and recorded, the doors are closed and sealed, and the seal and container numbers are noted. From the gate of the processor's cold store to the terminal, and on board the vessel, the reefer runs plugged into an external power supply; at sea it runs on the ship's power, and its temperature is monitored and logged throughout the voyage.

What would a careless loading have done? Suppose the cartons had arrived warm and been packed tight to the ceiling, blocking the airways. The unit would struggle to extract heat from a mass of warm cargo, ice would build on the evaporator and choke the airflow, the cargo temperature would drift upward, and the tuna would arrive softened and rejectable, with the temperature log telling the story of exactly when the cold chain failed.

Key lesson: a reefer maintains temperature; it does not rescue cargo loaded wrong. Pre-cooling, correct set point and clear airflow are the shipper's work, and the log will prove whether it was done.

WORKED EXAMPLE

A Badly Stuffed Container: What Tema Teaches

An importer in Tema buys a forty-foot container load of machine spare parts from a supplier in Shanghai, sold on terms that make the supplier responsible for arranging the export carriage. The supplier engages a casual stuffing team, which packs the box in a single afternoon. The container leaves within its weight limits and with a valid verified gross mass declared. The problems are all inside the doors.

Step 1: The stuffing errors. The team loads the heaviest cartons, filled with steel components, last and on top, because they were stacked nearest the warehouse door. Light cartons of filters and gaskets go in first, on the floor. The cargo is stacked to leave a wide, unbraced gap just inside the doors, and no dunnage, straps or bracing fill or restrain anything. All of this is invisible once the doors close and the seal goes on.

Step 2: The voyage. From Shanghai to Tema the ship rolls in a swell and the container is handled by cranes and chassis. Unrestrained loads obey physics: the unbraced gap lets the whole mass shift toward the doors, and the heavy steel cartons ride up and over, crushing the light cartons beneath them.

Step 3: The arrival. At Tema the seal is intact and matches the documents, so nothing was opened in transit. But when the de-stuffing team breaks the seal and opens the doors, a

collapsed wall of cargo greets them, a genuine safety hazard, and behind it they find crushed cartons, torn packaging and damaged components.

Step 4: Sorting out responsibility. The carrier received a sealed box in apparent good order and delivered it the same way; the damage did not happen to the outside of the box. A cargo survey traces the damage to improper packing and inadequate securing inside the container, which is the shipper's side of the operation, so the loss lands on the parties to the sale, not on the carrier, and the importer's relationship with its customer suffers while replacements are manufactured.

Who learns what? The supplier learns that stuffing discipline is not optional: heavy cargo low, light cargo high, every gap braced, dunnage used, the load spread evenly. The importer learns to specify packing standards in its purchase contracts and to consider loading supervision, because the box that arrived told the truth only when it was opened.

Key lesson: a container protects cargo from the outside world only after someone has protected the cargo from itself.

IMPORTANT NOTE

A Sealed Container Is Not an Inspected Container

The seal teaches an important and uncomfortable distinction: a container is sealed but not inspected. Fixing a seal does not mean anyone has checked what is inside, and no seal makes the contents lawful or accurate.

Responsibility for the contents rests with the party who stuffed the box. Customs examines declarations against goods; when a sealed container is selected for inspection and opened, the officers compare what they find with what was declared. If the cargo does not match the declaration, in quantity, description or anything else, the discrepancy is the declarer's problem, whoever physically loaded the cartons, because the shipper contracts and supervises the stuffing. A seal proves the doors stayed shut; it says nothing about whether what went in was right.

The practical disciplines follow. Supervise or trust only competent stuffing. Count what goes in and check it against the packing list before the doors close, because afterwards the box may travel for weeks. Record the container and seal numbers accurately. Declare against the real contents. And remember the arrival-side mirror of the same rule: on de-stuffing, check what comes out before signing for it, because the seal's evidence is only as useful as the receiver's diligence in comparing, recording and reporting what it shows.

IMPORTANT NOTE

Payload Is a Limit, Not a Suggestion

The weights taught in this lesson form a chain that must never be broken. Max gross is the rated maximum for container plus cargo. Tare is the empty container's own weight. Payload is what

remains for cargo: max gross minus tare, shown for each individual container on its equipment plate. The figures quoted in this lesson are typical design values; the plate on the actual box is the authority for that box.

Two refinements keep this honest. First, newer containers built and tested to the amended ISO 668 standard may carry a higher max gross rating, up to 36,000 kilograms, on their plates, while the traditional design rating of 30,480 kilograms remains the reference taught here; whichever rating applies, it is the plate that applies. Second, in practice the road weight limits of the countries on the route usually cap the real load before the container's rating does: a box legally able to carry 28 tonnes is useless if the truck carrying it may not.

Exceeding payload is not a negotiable corner-cutting. An overloaded container is refused at gates, must be reworked at cost and delay, endangers truck braking and crane lifting, and sits outside what vessel lashing plans assume. And the discipline closes with the verified gross mass: the shipper declares the actual verified weight of the packed box before sailing, so overloading is not merely risky, it is documented.

IMPORTANT NOTE

LCL Means Sharing a Container and a Fate

An LCL consignment does not travel alone, and neither do its problems. The consolidation taught in Lesson 5.3 puts your cargo in a box with other people's goods, and the whole box moves as one unit through every cut-off, gate and inspection on the journey. A single problem in one consignor's cargo, a late delivery to the CFS, incomplete shipping instructions, an undeclared dangerous good discovered at handling, can hold the entire container, so that everyone's cargo rolls to a later vessel together, exactly the rollover consequence Lesson 7.1 described at the cut-offs.

Prudent shippers therefore choose their consolidation arrangements deliberately. Use a reputable consolidator or forwarder with sound handling and documentation discipline, and ask about its CFS cut-off times rather than discovering them. Deliver your own cargo to the CFS early and complete, so your consignment is never the reason a box rolls. Keep communication active during the journey, as Lesson 5.4, Shipment Tracking and Customer Communication, argued, because when a consolidation is delayed you want to know early enough to manage the customer.

Finally, pack for the neighbours you cannot choose. Sturdy outer packaging, moisture protection and odour-proof wrapping for sensitive goods, and honest declaration of what your cargo is, so that the consolidator can plan compatibility, are how a small shipment survives a shared box. In LCL, your cargo's journey depends partly on strangers; pack, choose and schedule accordingly.

Practice questions (Lesson Checks)

Practice Question 1

An exporter in Kumasi has six pallets of retail-packaged cosmetics ready for a buyer in Hamburg. The cartons are attractively printed and easily crushed, the exporter wants to pack immediately after the production line finishes this week, and the forwarder says the LCL share price plus CFS charges would be similar in total to the price of an exclusive twenty-foot container.

Which option best explains why the exporter chooses FCL?

1. FCL is always cheaper than LCL for any shipment that fills more than half a container, regardless of handling or timing considerations.

2. FCL gives the shipment exclusive use of a sealed box: the cargo is stuffed once at the exporter's own premises, suffers no consolidation handling, and can be packed when production finishes rather than delivered to a CFS by an earlier cut-off. ✓

3. LCL shipments are not accepted on vessels serving European ports, so exporters of fragile goods have no choice but to book full containers.

4. FCL containers are inspected and approved by the carrier at stuffing, so exclusive use removes the exporter's responsibility for how the cargo is packed.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The lesson's decision factors are volume against a full box, cost per unit, handling risk, timing and cargo sensitivity. Here the volumes are middling and the costs are close, so the deciding factors are the qualitative ones. FCL means the six pallets are loaded once, at the exporter's convenience after production, into a box carrying no strangers' cargo, sealed by the exporter and untouched until destination. For easily crushed retail cartons and a tight production schedule, that combination of less handling and control of timing is exactly what FCL exists to provide.

Answer 1 overstates the cost logic: FCL is not always cheaper, and this scenario says the prices are similar, so cost does not decide it either way. Answer 3 invents a rule that does not exist; LCL is a normal service on European lanes, and the choice between it and FCL is a commercial decision. Answer 4 reverses a warning this lesson made explicitly: the carrier receives a sealed box and does not inspect or approve the stuffing, and responsibility for how the cargo is packed remains with the shipper.

The takeaway: when costs are comparable, handling risk and timing control are what justify booking the whole box.

Practice Question 2

A loading crew at a cold store is preparing a reefer shipment of frozen prawns for export. The cartons arrive from the chiller at ambient temperature because the store was busy, and the crew plans to pack them tightly to the ceiling, reasoning that the reefer will freeze them down to the set point during the voyage anyway.

Which statement best describes correct reefer practice?

1. The reefer is designed to maintain a set temperature, not to cool a warm load down to it, so the cargo must be pre-cooled to carriage temperature and loaded with clear airflow paths around it, and a warm, tightly blocked load risks temperature drift and a failed cold chain. ✓

2. The reefer unit will extract the heat from the cargo within a few hours of sailing, so loading temperature has no effect on the condition of the cargo at destination.

3. Packing cartons tightly against the refrigeration unit and to the ceiling is good practice in reefer containers, because a full box keeps the cold air in and reduces the unit's workload.

4. The correct response to a warm load is to lower the set point well below the specified temperature for the first part of the voyage, which compensates for the loading temperature without affecting the cargo.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

This lesson's reefer example made the principle explicit: a refrigerated container holds temperature, it does not rescue cargo loaded wrong. Cargo must enter the box already at its carriage temperature, which means pre-cooling the container to the set point and loading pre-cooled cargo quickly, and the cartons must be stacked so chilled air can circulate around them, with floor channels clear and nothing blocking the airflow from the unit. A warm load forces the unit to extract heat from the cargo itself, ice builds on the machinery, airflow chokes, and the cargo temperature drifts upward, exactly the failure pattern the frozen tuna example traced.

Answer 2 assumes the impossible: extracting the heat of a full warm load overwhelms the unit, and the temperature log will record the drift. Answer 3 blocks the very airflow the box depends on; tight, ceiling-high packing is the error, not the practice. Answer 4 invents a compensation that reefer discipline does not allow, since the set point is specified for the cargo and tampering with it creates its own quality and claim problems.

The takeaway: the cold chain is protected at stuffing, by pre-cooled cargo, a correct set point and clear airways, or it is not protected at all.

Practice Question 3

A logistics officer in Tema is supervising the stuffing of a forty-foot container with dense machinery components. The box is only half full by volume, but the cargo already weighs close to the payload shown on the equipment plate. A colleague says there is plenty of space left, so at least as much cargo again can safely be added.

Which statement best describes the correct position?

1. Volume is the only limit that matters in container loading, so any cargo that physically fits inside the box may be added provided the doors can still be closed.

2. For dense cargo the payload is the binding limit: the container will reach its rated maximum for cargo weight long before it is full, the actual load must respect the plate's rating and the road weight limits on the route, and the verified gross mass must reflect the true weight. ✓

3. The ISO amendment allows modern containers to carry up to 36,000 kilograms, so double the current cargo can be loaded into any forty-foot container without further checks.

4. Weight limits are the carrier's concern at the terminal gate, so the stuffing team should fill the remaining space and let the shipping line refuse the box if there is a problem.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The lesson's central rule of container selection is that a forty-foot box doubles the volume but not the permitted weight, and dense cargo is exactly the case where payload, not volume, binds. The payload is max gross minus tare, shown on that individual container's equipment plate, and it is a legal and safety limit, not a suggestion: exceeding it endangers truck braking, crane lifting and vessel lashing, and in practice the road weight limits on the route often cap the load below even the plate's figure. The declared verified gross mass must state the true weight of the packed box, so overloading cannot be hidden at sailing.

Answer 1 confuses the two dimensions of capacity, which is precisely the misconception the colleague has voiced. Answer 3 misuses the amendment: higher ratings apply to specific plating-tested containers on their own plates, not to any box, and they do not remove the road limits. Answer 4 abandons the shipper's responsibility; the weight is declared by the shipper, and a refused or reworked container costs delay and money at the worst moment.

The takeaway: check kilograms against the plate and the road, and let volume guide the choice of box, never the final amount of dense cargo stuffed into it.

Fact-check appendix (audit notes — not for entry)

Definitions of containerisation, shipping container, FCL, LCL, TEU, FEU, dry container, high-cube container, refrigerated container, open-top container, flat-rack container, container seal, container number, container freight station, cargo securing and verified gross mass follow the lesson brief 7.3 key terms. — source: lesson brief 7.3, Key Terms (curriculum database).

Freight containers are standardised under ISO 668 (classification, dimensions, ratings) and ISO 1496 (specification and testing). — source: research sheet trade-transport-standards.md, section 6 (containers, ISO 668/ISO 1496).

Typical 20-foot dry container: internal approximately 5.898 x 2.352 x 2.393 metres (about 33 cubic metres); max gross mass 30,480 kg; tare approximately 2,230 kg; payload approximately 28,250 kg. Typical 40-foot dry container: internal approximately 12.032 x 2.352 x 2.393 metres (about 67.5 cubic metres); max gross commonly rated 30,480 kg (some 32,500 kg); tare approximately 3,750-3,800 kg; payload approximately 26,700 kg. — source: research sheet trade-transport-standards.md, section 6.

40-foot high-cube: external height 9 ft 6 in (2.896 m), internal height approximately 2.698 m, volume approximately 75-76 cubic metres, footprint identical to standard 40-foot. — source: research sheet trade-transport-standards.md, section 6.

Tare is the empty container weight; payload is the maximum permissible cargo weight (max gross minus tare); max gross is the safety-rated total of container plus cargo; the equipment plate is authoritative per box. — source: research sheet trade-transport-standards.md, section 6 (terminology) and DO NOT CLAIM item 6 (payload derived from tare per equipment plate).

The amended ISO 668 standard raised the permissible maximum gross mass to 36,000 kg for plating-tested containers; in practice road weight limits, not the container rating, usually cap payload; 30,480 kg is the traditional design maximum; no amendment year is stated in the lesson. — source: research sheet trade-transport-standards.md, section 6 and DO NOT CLAIM item 6.

Reefer containers are about 6 per cent of the fleet; ISO tank containers hold liquids, gases and powders, often dangerous goods, and are under 1 per cent of the fleet; open tops have a removable tarpaulin roof for crane-loaded over-height cargo; flat racks have no sides or roof and carry out-of-gauge machinery; 1 FEU equals 2 TEU and a 40-foot high cube still counts as 1 FEU. — source: research sheet trade-transport-standards.md, section 6.

The shipper must obtain and communicate the verified gross mass of every packed container before loading (SOLAS VI/2, mandatory since 1 July 2016); no container without VGM may be loaded. — source: research sheet trade-transport-standards.md, section 7 (ocean freight operations).

Cargo and VGM cut-offs fall before sailing and missing a cut-off usually means rollover to the next vessel; CFS charges apply to container freight station handling of LCL cargo. — source: research sheet trade-transport-standards.md, section 7.

Minus 18 degrees Celsius (0 degrees Fahrenheit) is a widely used carriage set point for frozen cargo; reefer best practice is to pre-cool the container and load pre-cooled cargo, keep airflow circulating around the cargo (floor channels clear, cargo below the load line), and maintain the set point throughout. — source: verified via live web research: TT Club "Perishable Cargoes" guidance and ONE "Refrigerated Cargo Handling Guideline", consistent with IMO informative material on transport of perishable cargo; set point in the worked example stated as the shipper's instruction.

Reefers run on external electrical power at the terminal and on the vessel's electrical supply at sea. — source: general industry practice, consistent with the reefer guidance above.

Seals are numbered, the number is recorded on shipping documents, and an intact seal matching the documents evidences that doors were not opened in transit; customs applies its own seals for cargo moving under customs control, for example in transit regimes. — source: general business knowledge for seal practice; customs seals in transit per research sheet trade-transport-standards.md, section 12 (ECOWAS ISRT, customs transit under seals).

FCL does not require a full box; it describes exclusive use of the container by one shipper or consignee; the container remains the carrier's equipment. — source: general business knowledge, consistent with the FCL/LCL definitions in lesson brief 7.3.

All quantities, weights, temperatures, volumes and route details in the worked examples (nine pallets, 4,800 kg, 19 cubic metres; minus 18 degrees Celsius set point; Tema and Shanghai consignments) are illustrative for teaching purposes. — source: WRITING_SPEC.md rule 5.1.

Lesson 7.4 — Shipping Charges, Freight Rates, and Surcharges

Summary

This lesson introduces the financial components associated with international cargo transportation. Learners will examine how freight rates are established, the differences between base freight charges and additional surcharges, and the factors influencing shipping costs. The lesson also explains common ocean and air freight charges, freight calculation methods, and the importance of reviewing quotations to identify the total cost of transportation.

Learning objectives

- Define freight rates, shipping charges, and surcharges.
- Identify the major components of ocean and air freight pricing.
- Explain the factors influencing international freight rates.
- Distinguish between base freight charges and additional shipping surcharges.
- Describe common charges associated with containerised and consolidated shipments.
- Explain how weight, volume, container size, and transportation routes affect freight calculations.
- Identify the differences between prepaid and collect freight arrangements.
- Interpret the basic components of a freight quotation or shipping invoice.
- Perform simple freight calculations using specified rates and shipment details.
- Recognise the importance of verifying shipping charges before confirming transportation arrangements.

Key terms

Freight Rate — The price charged for transporting cargo according to an applicable pricing basis.

Base Freight — The principal transportation charge before separately stated surcharges or additional services.

Freight Surcharge — An additional charge applied to a freight service to account for specified costs or operating conditions.

Ocean Freight Charge (OFC) — The charge for transporting cargo by sea under the applicable freight arrangement.

Air Freight Charge — The charge for transporting cargo by aircraft, generally calculated using the applicable chargeable weight and tariff or agreed rate.

Bunker Adjustment Factor (BAF) — A maritime freight adjustment or surcharge associated with changes in fuel costs, where applicable under the carrier's pricing structure.

Currency Adjustment Factor (CAF) — A freight adjustment or surcharge associated with exchange-rate fluctuations, where applicable.

Terminal Handling Charge (THC) — A charge associated with specified cargo or container handling activities at a terminal.

Documentation Fee — A charge for preparing, issuing, or processing specified transportation or shipping documents.

Chargeable Weight — The weight used to determine freight charges according to the applicable transportation pricing rules.

Volumetric Weight — A calculated weight based on cargo dimensions and an applicable volume-to-weight conversion factor.

Weight or Measurement (W/M) — A freight rating method in which charges are calculated using weight or volume, generally whichever produces the higher charge under the applicable tariff.

Freight Prepaid — A freight payment designation indicating that the relevant freight charges are payable at or before shipment according to the agreed arrangement.

Freight Collect — A freight payment designation indicating that the relevant freight charges are payable at destination or by the designated receiving party under the agreed arrangement.

All-In Freight Rate — A quoted transportation rate that includes the charges expressly covered by the quotation, which may exclude other services or destination expenses.

Freight Invoice — A document issued to request payment for transportation and associated services.

Teaching content

KEY CONCEPT

Freight Rates, Shipping Charges, and Surcharges

Every international shipment carries two loads: the cargo, and the charges that come with moving it. This lesson teaches the language of those charges and the practical skill of reading a freight bill from its first line to its last.

Three terms anchor everything that follows.

1. Freight rate. The freight rate is the price charged for transporting cargo according to an applicable pricing basis. It is a price per unit of transport: so much per container, per tonne, or per kilogram. The pricing basis, meaning what the rate is charged on, is the single most important fact about any rate, and the next concept examines it.

2. Shipping charges. Shipping charges are the total amounts payable for transporting the cargo and for the related services around it: handling, documentation, security, terminal work, delivery. The freight rate is one line among these charges, usually the largest, but never the only one.

3. Freight surcharge. A freight surcharge is an additional charge applied to a freight service to account for specified costs or operating conditions, such as the price of fuel or the pressure of a busy season. Surcharges are stated separately from the base rate precisely because they can change when the underlying cost or condition changes.

Beginners read a freight quotation down to the base rate and stop. Professionals read it as three layers: the base freight, the principal transportation charge before separately stated surcharges or additional services; the surcharges attached to the main carriage; and the local service charges raised at the origin and the destination. The whole of this lesson builds the habit of seeing all three layers, because the gap between the first layer and the last is where costly surprises live.

KEY CONCEPT

The Pricing Basis: What the Rate Is Charged On

A freight rate is never a single sum invented for one shipment. It is a price per unit, and the unit, the pricing basis, is stated or implied in every quotation and every tariff. Three pricing bases dominate international freight.

1. Per container, for FCL ocean cargo. Lesson 7.3, Containerisation: FCL, LCL, and Container Types, explained that a full container load is the exclusive use of a box by one shipper. Ocean rates for such containerised cargo are commonly quoted per container: one price for the movement of the box, whether the shipper has filled it or not. A twenty-foot container and a forty-foot container are different pieces of equipment, so they are priced as different items; the larger box is priced higher than the smaller, typically well below double, because it is still one unit lifted once. The TEU and FEU counting units taught in Lesson 7.3 are the industry's currency for this kind of pricing.

2. Per weight or measurement tonne, for LCL ocean cargo. A less than container load pays a share of a box, and the share is rated by weight or measurement (W/M), a freight rating method in which charges are calculated using weight or volume, generally whichever produces the higher charge under the applicable tariff. Weight is taken in metric tonnes and measurement in

cubic metres; the greater of the two becomes the quantity charged, a unit often called the revenue tonne. Light, bulky cargo therefore pays on space, and dense cargo pays on weight.

3. Per kilogram of chargeable weight, for air cargo. Air freight is charged on chargeable weight, the greater of the actual gross weight and the volumetric weight, the volumetric weight being a calculated weight based on cargo dimensions and an applicable volume-to-weight conversion factor, as Lesson 7.2, Air Cargo Operations and Cargo Acceptance, taught in full.

The same consignment of goods can carry three different price tags on three different bases, which is why the pricing basis must always be identified before any comparison is made.

KEY CONCEPT

Base Freight: The Principal Transportation Charge

Base freight is the principal transportation charge before separately stated surcharges or additional services. It is the price of the main carriage itself: the vessel's or the aircraft's work of carrying the cargo across the ocean or along the air lane. On the sea leg, the base line-haul charge is commonly called the ocean freight charge, the charge for transporting cargo by sea under the applicable freight arrangement.

How are base rates established? Lesson 7.1, Ocean Freight Operations and Vessel Schedules, described liner shipping: scheduled services on fixed routes selling space by the container. Liner base rates emerge from published tariffs, from contracts between carriers and large shippers, and from continuous market negotiation in between. There is no universal price list; each carrier prices each trade lane according to its own costs and commercial position, which is why the same container can attract different base offers from different lines in the same week.

What does the base rate reflect? Several influences are visible even before surcharges are considered.

1. The trade lane. A route with heavy competition and abundant capacity is priced differently from a thin route served by few services. Distance matters, but the balance of trade on the lane often matters more.

2. The equipment. A twenty-foot box, a forty-foot box, a high cube and a reefer are priced as different items, and specialised equipment carries a different base from a dry box.

3. The service pattern. A direct sailing and a routing with a transhipment are different products, and the base rate reflects the difference.

4. The cargo itself. For cargo rated by weight or measurement, the consignment's own figures decide the quantity charged.

The base rate, in other words, is the foundation of the bill but not the whole building. The next two concepts add the storeys above it: surcharges, and the local charges at each end.

KEY CONCEPT

What Moves Freight Rates

Freight rates are prices, and prices move. Understanding the main forces behind the movement turns a freight quotation from a mystery into a readable market signal.

1. Fuel costs. Vessels and aircraft burn fuel by the tonne, and fuel is one of the largest costs of the main carriage. When fuel prices rise sharply, carriers do not usually rewrite every base rate; they recover the movement through fuel surcharges, as the next concept explains.

2. Exchange rates. Carriers earn in one currency and pay in others, so exchange-rate movements feed into pricing on international lanes, again often through a stated adjustment.

3. Capacity and demand on the lane. When trade on a route is booming and sailings are full, rates rise; when capacity exceeds cargo, they soften. Seasonality drives this rhythm: harvest exports, festive retail peaks and other recurring surges create predictable pressure periods, and a peak-season surcharge is one of the ways carriers manage them.

4. Route and service conditions. Congestion at a major port, security conditions on the route, and the choice between a direct service and one with a transhipment all shape what a lane costs to serve.

5. The cargo and its equipment. Weight, volume, stowage characteristics, the need for a reefer or other special equipment, and the extra work that dangerous goods require all influence the price calculated for a specific shipment.

Two lessons follow for the practitioner. First, a rate quoted today is a snapshot of these forces today; the forces keep moving, which is why quotations carry validity periods, a point this lesson returns to. Second, because many of the forces act through surcharges rather than the base rate, a shipper who watches only headline base movements reads only half the market. No amounts are quoted in this lesson, and none can be: the figures change continuously, and every carrier's tariff is its own.

KEY CONCEPT

Surcharges: Why the Bill Has Extras

A freight surcharge is an additional charge applied to a freight service to account for specified costs or operating conditions. The logic is simple: the base rate is relatively stable between reviews, while the costs behind it, fuel, currency, seasonal pressure, swing from month to month. Rather than rebuilding the whole tariff every time conditions move, carriers attach separately stated surcharges to the main carriage and adjust those instead. The surcharges below are taught as concepts; this course quotes no amounts for them, because each carrier's surcharges are tariff items of its own and they change continuously.

1. Bunker Adjustment Factor (BAF). The bunker adjustment factor is a maritime freight adjustment associated with changes in fuel costs, where the carrier's pricing structure provides for it. Bunker is the traditional word for a ship's fuel. A BAF rises and falls with the cost of bunkers, and many carriers now express the calculation through index clauses published with the rate, so the surcharge moves by stated rule rather than by whim.

2. Currency Adjustment Factor (CAF). The currency adjustment factor is a freight adjustment associated with exchange-rate fluctuations, where applicable. It compensates the carrier for the gap between the currency it earns in and the currencies it pays in.

3. Peak-season and congestion surcharges. A peak-season surcharge applies during recurring high-demand periods on a lane; a congestion surcharge recovers the cost of delays and crowding at a busy port. Both are tied to operating conditions, not to the cargo itself.

4. The general rate increase (GRI). The GRI is the liner industry's instrument for lifting base rates themselves: an announced, across-the-board upward adjustment of base freight rates on a trade lane, typically timed for contract seasons or capacity management. It is distinct from ad-hoc surcharges, because a GRI changes the foundation, the base rate, rather than adding a line above it.

The practitioner's rule follows from this anatomy: a surcharge attached to the carriage can change between quotation and shipment, so no surcharge figure should ever be assumed to have frozen.

KEY CONCEPT

Local Charges: The Long Tail at Origin and Destination

Beside the main carriage and its surcharges sits the third layer: local service charges, raised at the ports, terminals, airports and offices through which the shipment passes. These are the charges beginners overlook and freight bills never forget. The commonest are the following.

1. Terminal Handling Charge (THC). The terminal handling charge is a charge associated with specified cargo or container handling activities at a terminal. In container shipping a THC is raised at each end: the origin THC covers the lifting work between road chassis, stacking position and vessel at the port of loading, and the destination THC covers the mirror-image work after discharge. They are separate charges from separate facilities, and a quotation that includes only one of them tells half a story.

2. Documentation fee. The documentation fee is a charge for preparing, issuing or processing transportation documents, most familiarly the issuance of the bill of lading.

3. Seal charge. The container seal taught in Lesson 7.3, the numbered security device fixed through the closed doors, is itself billed as a small line item.

4. ISPS security fee. Maritime and port security measures are required by international convention, under the International Ship and Port Facility Security Code within the framework of the Safety of Life at Sea Convention, and carriers and terminals recover their cost through a security charge on the cargo.

5. CFS charges. For LCL cargo, the container freight station work described in Lesson 7.3, receiving consignments, stuffing them into the consolidation box, and later deconsolidating them at destination, is billed as CFS charges at both ends.

6. Collection and delivery charges. The road carriage that brings cargo to the port or airport of loading, or takes it from the port or airport of discharge to the premises, is a service of its own, charged separately unless a door-to-door scope bundles it in.

The practical lesson is arithmetic: door-to-door cost equals the freight plus a long tail of local charges, and the tail is not trivial. Destination charges are also commonly payable by the consignee, in the local currency at the far end, which is exactly why an origin-side port-to-port quotation can look strikingly small next to what the shipment actually costs to land at the door.

KEY CONCEPT

Prepaid and Collect: Who Pays, and Where

Every freight bill must be paid by someone, at some end. The two designations that record this are among the most consequential markings on a transport document.

1. Freight prepaid. A freight prepaid designation indicates that the relevant freight charges are payable at or before shipment. In practice the shipper, or another party on the origin side, settles the freight before the goods sail, and the transport document is marked to show it.

2. Freight collect. A freight collect designation indicates that the freight charges are payable at destination, or by the designated receiving party, under the agreed arrangement. The consignee at the far end settles the charges as a condition of taking delivery.

Which arrangement applies is a commercial decision, and it usually follows the sales contract. Lesson 4.3, Incoterms and Allocation of Responsibilities, explained that under Incoterms 2020 the E and F rules place the arrangement of the main carriage with the buyer, while the C and D rules place it with the seller. Where the seller arranges the carriage, freight is commonly prepaid; where the buyer arranges it, the movement is often booked on collect terms so that charges are recovered at the destination end. The parties can agree otherwise, so the marking on the document, not the assumption, is the authority.

The operational consequences are real. On collect terms the consignee must be ready to pay the freight, and commonly the destination local charges as well, promptly on arrival, because carriers release cargo against settlement. An exporter who ships collect to a consignee who is slow to pay, or unreachable, has engineered a delay for cargo it no longer controls. Prepaid or

collect is therefore verified before booking, marked consistently on the documents, and communicated to everyone in the chain.

KEY CONCEPT

Reading a Freight Quotation and a Freight Invoice

Lesson 5.2, Freight Quotations and Shipment Booking, taught how quotations are requested, structured and compared. This lesson adds the reading skill that turns that knowledge into cost control: knowing what each part of the document is telling you. A freight quotation or shipping invoice, however it is laid out, contains the same anatomy.

1. The parties and the shipment reference. Who is quoting, for whom, and for which movement: places, equipment, cargo description.

2. The scope of service. Port to port, port to door, or door to door. The scope is the boundary of the price; everything outside it is an exclusion, however small the print.

3. The pricing basis and quantity. One twenty-foot container, a number of revenue tonnes under W/M, or a chargeable weight in kilograms. This line lets you verify how the price was built.

4. The base freight line. The principal transportation charge, on the stated basis.

5. The surcharge lines. Fuel, security and their fellows, each stated separately.

6. The origin local charges. Terminal handling, documentation, seal and origin handling within scope.

7. The destination charges. Often the most important lines, because they are frequently shown as excluded, or as payable locally by the consignee.

8. The conditions. Currency, payment basis, prepaid or collect, and the validity date, the date until which the quoted prices hold.

Two key terms complete the anatomy. An all-in freight rate is a quoted transportation rate that includes the charges expressly covered by the quotation, which may exclude other services or destination expenses; all-in, in other words, means read the list of what is included, never assume that nothing else can be charged. A freight invoice is a document issued to request payment for transportation and associated services; it is the quotation's post-shipment counterpart, and the professional habit is to reconcile each invoice line against the quotation and the booking before paying, querying any line that was never quoted.

KEY CONCEPT

Air Freight Charges in Parallel

Everything this lesson has taught about layers applies unchanged in the air. The names differ; the anatomy does not.

1. The air freight charge. The air freight charge is the charge for transporting cargo by aircraft, calculated using the applicable chargeable weight and the tariff or agreed rate. Lesson 7.2 taught chargeable weight: the greater of the gross weight and the volumetric weight, with volumetric weight computed from dimensions, under the IATA standard by dividing length times width times height in centimetres by 6,000. The pricing basis is therefore kilograms of chargeable weight, and the base freight line is that weight multiplied by the applicable rate.

2. Air surcharges. Airlines attach fuel surcharges, responding to the cost of jet fuel, and security surcharges, recovering the cost of screening and security regimes, to the carriage. Both are commonly stated per kilogram, so they scale with the chargeable weight exactly as the base does.

3. Air local charges. Airports have their own tail: cargo terminal charges for handling and screening at the origin airport, mirror-image terminal charges at the destination, documentation and air waybill fees, and collection or delivery carriage at either end.

The air waybill itself usually serves the freight-bill function on the air leg: Lesson 7.2 described the AWB as receipt for the goods, evidence of the contract of carriage and freight bill, and its charges often appear on its face. Where a forwarder quotes an all-in price per kilogram, the same caution as at sea applies: an all-in rate covers only the charges expressly listed, so the list is read before the price is believed.

The transferable conclusion deserves stating plainly. Whatever the mode, the total cost of a shipment is the main carriage, plus the surcharges attached to it, plus the local charges at the ends. A learner who can read that structure on an ocean bill can read it on an air bill.

KEY CONCEPT

Doing the Arithmetic: Simple Freight Calculations

Freight arithmetic rewards method over speed. All figures in this concept are illustrative for teaching purposes only. Three model calculations cover the common cases.

1. FCL ocean freight. Multiply the base rate per container by the number of containers, then add surcharges and local charges. Example: two twenty-foot containers at a base of 1,200 US dollars each give a base freight of 2,400; adding surcharges of 350 and origin local charges of 300 gives an origin-side total of 3,050 US dollars. Destination charges, if excluded, are added separately to see the door cost.

2. LCL freight under W/M. Convert the consignment to both units: weight in metric tonnes, volume in cubic metres. A consignment of 2,400 kilograms and 6.8 cubic metres is 2.4 tonnes against 6.8 cubic metres; W/M charges the higher, so the chargeable quantity is 6.8 revenue tonnes. At an illustrative rate of 85 US dollars per revenue tonne, the freight is 6.8 multiplied by 85, which is 578 US dollars, before CFS and other local charges are added.

3. Air freight. Establish the chargeable weight, multiply by the rate per kilogram, then add per-kilogram surcharges and flat local charges. A consignment of 40 chargeable kilograms at 4 US dollars per kilogram carries a base freight of 160; surcharges of 48 and local charges of 112 lift the bill to 320 US dollars, a structure the final worked example examines in full.

The method is always the same four steps.

1. Identify the pricing basis: per container, per revenue tonne, or per chargeable kilogram.

2. Compute the quantity: the containers, the greater of weight and measurement, or the greater of gross and volumetric weight.

3. Multiply the quantity by the base rate.

4. Add each surcharge and local charge in its turn, and state the currency.

Perform the steps in order and the total explains itself, line by line, exactly as a freight invoice should.

WORKED EXAMPLE

Reading the Freight Invoice: One Twenty-Foot Container from Tema

A processor of cocoa products in Tema exports a twenty-foot container of cocoa butter in cartons to a buyer in Hamburg, booked with a forwarder on a liner service as a full container load, port to port, freight prepaid. When the vessel sails, the forwarder's freight invoice arrives. All figures in this example are illustrative for teaching purposes.

Step 1: Confirm the frame. The invoice names the parties, the one twenty-foot container, the route from Tema to Hamburg, and the scope: port to port. The pricing basis is per container, so no weight or volume arithmetic applies. The payment basis reads freight prepaid.

Step 2: Read the main-carriage lines. The base ocean freight appears first: 1,150 US dollars for the twenty-foot container. Below it stands the bunker adjustment factor, the fuel-related surcharge: 180 US dollars. Main carriage subtotal: 1,330 US dollars.

Step 3: Read the origin local lines. The origin terminal handling charge appears next at 150 US dollars, then the documentation fee for issuance of the bill of lading at 60, the container seal at 15, and the ISPS security fee at 25. The origin locals total 250 US dollars.

Step 4: Reconcile and total. The exporter checks each line against the quotation accepted weeks earlier: the base matches, the BAF figure is within the terms quoted, the documentation fee covers one bill of lading, and no unexplained lines appear. The invoice total is 1,580 US dollars, marked freight prepaid, settled at origin.

Step 5: Add the lines that are not on the invoice. Because the scope was port to port, the destination work is absent. At Hamburg the buyer's agent pays the destination terminal

handling charge of 220 US dollars and delivery to the buyer's warehouse of 190, a further 410 US dollars settled locally. The true door-to-door cost of the movement is therefore 1,580 plus 410: 1,990 US dollars.

What does this example demonstrate?

A freight invoice is read as three layers: base freight, surcharges, and local charges. It also covers only the scope booked, so the invoice that looked like the whole cost was, on a port-to-port movement, only the origin-side part.

Key lesson: reconcile every line against the quotation before paying, and always add the destination tail before calling a shipment cheap or dear.

WORKED EXAMPLE

Two Quotations, One Container: Comparing Scope in Hamburg

A confectionery manufacturer in Hamburg buys a twenty-foot container of cocoa products from a processor in Tema. The agreed commercial terms make the buyer responsible for arranging the main carriage, so the manufacturer's logistics desk invites two forwarders to quote for moving one twenty-foot container from Tema to the company's warehouse in Hamburg. All figures in this example are illustrative for teaching purposes.

Step 1: Quote A, port to port. Forwarder A quotes 1,575 US dollars: base ocean freight 1,180, bunker adjustment factor 175, origin terminal handling charge 150, documentation fee 50, and container seal 20. The quotation is valid for seven days, and its conditions list the exclusions: destination terminal handling, assistance with import clearance, and delivery to the warehouse, all payable locally in Hamburg by the consignee.

Step 2: Quote B, door to door. Forwarder B quotes 2,040 US dollars for the same box on comparable sailing options: base ocean freight 1,150, bunker adjustment factor 175, origin local charges 210 covering terminal handling, documentation and seal, destination terminal handling charge 230, assistance with import clearance 90, and delivery to the warehouse 185. The quotation is valid for fourteen days.

Step 3: Rebuild to equal scope. The desk adds to Quote A the very charges it excludes, taking the figures from Quote B's stated lines: destination terminal handling 230, clearance assistance 90, delivery 185, a destination tail of 505. Quote A's like-for-like door-to-door cost is therefore 1,575 plus 505: 2,080 US dollars. Against Quote B's 2,040, the dearer-looking quotation is in fact 40 US dollars cheaper on equal scope, and it carries twice the validity period.

Step 4: Decide. The desk recommends Quote B: the lower equal-scope total, one responsible party for the whole movement, and a validity period long enough for the production plan behind the order.

What does this example demonstrate?

Headline totals built on different scopes cannot be compared. The comparison only means something when both quotations are rebuilt to the same scope, and the charges one quotation pushes to the destination end are real money however they are labelled.

Key lesson: compare like with like, first the total scope, then the price, then the conditions.

WORKED EXAMPLE

The Air Bill That Doubled: A Bulky Carton from Accra

An exporter of hand-woven baskets and other home decoration items in Accra flies one large carton to a retail buyer in London, departing from Kotoka International Airport. The carton measures 80 by 60 by 50 centimetres and weighs 26 kilograms. The exporter, thinking in kilograms, expects a modest bill. All figures in this example are illustrative for teaching purposes.

Step 1: Establish the chargeable weight. Applying the rule Lesson 7.2 taught, the volumetric weight is 80 multiplied by 60 multiplied by 50, which is 240,000, divided by the IATA divisor of 6,000: 40 kilograms. The gross weight is 26 kilograms. Chargeable weight is the greater of the two: 40 kilograms. The carton will pay for the space it occupies, not for what it weighs.

Step 2: Compute the base freight. The agreed rate is 4 US dollars per chargeable kilogram, so the base air freight charge is 40 multiplied by 4: 160 US dollars. Even this correct figure already exceeds the exporter's mental estimate, which had multiplied the real weight of 26 kilograms by the rate.

Step 3: Add the surcharges. The fuel surcharge of 0.80 US dollars per kilogram adds 32; the security surcharge of 0.40 per kilogram adds 16. The surcharges together add 48 US dollars, all scaling with the chargeable weight.

Step 4: Add the local charges. Origin terminal and screening charges at Accra add 40; the air waybill and documentation fee adds 24; at London, destination terminal charges add 28 and delivery to the buyer's premises adds 20. The local tail totals 112 US dollars.

Step 5: Total the bill. Base 160, surcharges 48, locals 112: the freight bill comes to 320 US dollars. The charges around the carriage, 160 US dollars of them, exactly match the carriage itself, and the total is roughly three times the exporter's first estimate.

What does this example demonstrate?

In air freight the base rate is only the floor of the bill. The chargeable weight rule raised the quantity charged from 26 to 40 kilograms, and the surcharges and terminal charges then doubled the apparent base rate.

Key lesson: budget the bill, not the rate, and give the forwarder exact dimensions and weight so the true total is quoted before anyone commits.

IMPORTANT NOTE

A Low Headline Rate Is Not a Cheap Shipment

The most expensive reading error in freight is stopping at the headline. A port-to-port quotation from which every local charge is excluded will almost always look cheaper than a door-to-door quotation that includes them, and the difference between the two totals is not saving; it is scope. Remember the definition of the all-in freight rate: it includes the charges expressly covered by the quotation, which may exclude other services or destination expenses. All-in promises a list, not an empty space for extras.

The discipline that prevents the error is rebuilding. Before two quotations are compared, each is taken down to its lines and rebuilt to the same scope: origin locals, main carriage, surcharges, destination locals, delivery. Only totals at equal scope are comparable, as the Hamburg example demonstrated, and only after the conditions, validity, currency and transit time, have also been read.

One asymmetry deserves particular care in Ghanaian trade and everywhere else: destination charges are commonly payable by the consignee at the far end. A low headline rate on the export side can quietly hand a large tail to the buyer, and an importer who accepts a collect arrangement without first pricing the destination end has bought a surprise at the port. Whether you are the shipper or the consignee, the question to ask of any quotation is not whether the number is low, but what is not on it.

IMPORTANT NOTE

Surcharges Move: A Quotation Lives Only Until Its Validity Ends

This lesson has repeatedly said that no surcharge figure can be assumed to hold. The reason is structural. The bunker adjustment factor moves with fuel, the currency adjustment factor with exchange rates, congestion and peak-season surcharges with operating conditions, and a general rate increase can lift the base rate of a whole trade lane at a stroke. The charges are designed to move; that is their purpose.

This is why the freight validity period taught in Lesson 5.2, Freight Quotations and Shipment Booking, matters so much in this lesson's world. A quotation holds only until its stated validity date, and the surcharges finally invoiced are those applicable when the shipment actually sails, not those printed on a document that has expired. Three habits convert this from a warning into a procedure.

1. Accept within validity. If the booking decision slips past the validity date, ask for the prices to be reconfirmed before instructing anyone to proceed.

2. Reconfirm after change. If the shipping date moves, or a carrier announces a rate adjustment on the lane, revalidate the quotation rather than hoping it still stands.

3. Reconcile at invoice. Check the invoiced surcharge lines against the quotation and the booking, and query promptly any line that was never quoted.

A quotation figure is a fact about a date. Carried beyond that date into a sales price or a budget without reconfirmation, it becomes the quietest kind of cost overrun.

IMPORTANT NOTE

What a Freight Bill Does Not Include

A freight bill, however long, is not a list of everything a shipment will ever cost, and the verification habit this lesson builds includes knowing what to look for elsewhere.

Customs duties and taxes never appear on a freight invoice, because they are statutory charges assessed by customs authorities, not prices charged by carriers or forwarders; the import charges due at the destination are the importer's separate business under the agreed commercial terms. Cargo insurance is likewise a separate service, arranged and charged separately unless a specific quotation expressly includes it. And one sentence this lesson owes to completeness: demurrage, detention and storage are separate charge families that begin when free time ends, and Lesson 9.4, Demurrage, Detention, and Storage Charges, treats them fully.

The practical conclusion is the one this lesson closes on. Before confirming any transportation arrangement, confirm three things in writing: the scope, so the exclusions are visible; the validity, so the prices are alive; and the exclusions, so the tail is expected. After the shipment, reconcile the freight invoice against the quotation and the booking before paying it, line by line, querying whatever was never quoted. A learner who can define a freight rate, name the three layers of a freight bill, mark a document prepaid or collect knowingly, and rebuild two quotations to equal scope already reads freight documents more carefully than the common mistakes of this trade assume.

Practice questions (Lesson Checks)

Practice Question 1

An importer in Takoradi is arranging shipment of machine parts from Shanghai. Forwarder A quotes 2,600 US dollars, port to port, and lists destination terminal handling, assistance with customs clearance, and delivery to the warehouse as excluded charges payable locally at destination. Forwarder B quotes 2,930 US dollars, door to door, including all of those charges. A colleague says Forwarder A is 330 US dollars cheaper and should be chosen immediately.

Which option best explains the correct way to compare the two quotations?

1. Forwarder A is cheaper, because the base freight and surcharges are the true cost of shipping, and local charges at destination are minor extras that never change the comparison.

2. The quotations cannot be compared at all, because every forwarder is free to quote whatever total it wishes and no common basis exists between offers.

3. Both quotations must first be rebuilt to the same scope by adding Forwarder A's excluded destination charges to its total, because the cheaper headline may cost more once the shipment actually reaches the door. ✓

4. Forwarder B should be chosen automatically, because a door-to-door quotation is always better value than a port-to-port quotation regardless of the figures.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The two quotations are built on different scopes: Forwarder A prices the main carriage and the origin charges only, while Forwarder B prices the whole movement including the destination tail. Until A's excluded destination charges are added to its total, the 330 US dollar difference measures scope, not value. Rebuilding both quotations to equal scope, exactly as the Hamburg worked example demonstrated, is the only comparison that means anything; Forwarder A may still prove cheaper after the rebuild, but nobody can know before it is done.

Answer 1 dismisses destination charges as minor extras, which is precisely the misconception this lesson dismantles: the local tail can rival the main carriage itself. Answer 2 abandons comparison altogether and forfeits the discipline of reading quotations line by line that Lessons 5.2 and 7.4 both teach. Answer 4 commits the mirror-image error, assuming that inclusiveness guarantees value; a door-to-door quotation can still be the dearer buy once both are rebuilt to equal scope.

The takeaway: compare totals at equal scope first, then weigh price, validity and service conditions.

Practice Question 2

A forwarder in Accra is rating an LCL consignment of plastic household goods for ocean shipment. The consignment weighs 2,400 kilograms, which is 2.4 tonnes, and measures 6.8 cubic metres. The LCL rate is quoted per weight or measurement tonne, whichever produces the higher charge.

Which statement best describes how the consignment will be charged?

1. On the weight of 2.4 tonnes, because LCL cargo is always charged on its actual weight regardless of the space it occupies.

2. On the measurement of 6.8, because under weight or measurement rating the charge follows whichever of the weight in tonnes and the volume in cubic metres produces the higher charge. ✓

3. On the total of 9.2, because the weight and the measurement are added together before the rate is applied.

4. On the number of packages, because consolidated cargo is charged per pallet rather than by weight or by volume.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Weight or measurement rating exists because a shipper's consignment can exhaust either the vessel's weight allowance or its space, so the tariff charges the greater of the two. The consignment is converted to both units, 2.4 tonnes against 6.8 cubic metres, and the higher figure, 6.8 revenue tonnes, becomes the quantity to which the LCL rate is applied. Light, bulky cargo such as plastic household goods is the classic case that pays on measurement, not weight.

Answer 1 assumes weight always governs, which would undercharge exactly the cargo that fills a box without weighing much; weight governs only when it is the greater figure. Answer 3 misreads the rule by adding the two measures together; they are alternatives, never combined, and adding them would charge the same cubic metre or kilogram twice. Answer 4 invents a per-package basis; consolidation rates are built on the consignment's weight or measurement, not on a pallet count.

The takeaway: under W/M, compute both the tonnage and the cubic metres, charge the higher, and never assume which one will win before the figures are converted.

Practice Question 3

A trading company in Accra ships a consignment of shea butter to a buyer in Dakar on freight collect terms. Before the goods sail, the logistics officer wants to be certain what the marking on the transport document means for payment.

Which statement best describes a freight collect arrangement?

1. The freight charges are payable at destination, normally by the designated receiving party, who is expected to settle them before the cargo is released. ✓

2. The freight charges are payable at or before shipment in Accra, and the transport document would instead carry the marking freight prepaid.

3. The freight charges may be settled by the consignee at any convenient time after taking delivery of the goods.

4. The freight charges are included in the price of the goods, so neither party pays the carrier separately.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

A freight collect designation indicates that the freight charges are payable at destination, or by the designated receiving party, under the agreed arrangement. Carriers release cargo against settlement, so the consignee in Dakar must be ready to pay the freight, and commonly the destination local charges as well, promptly on arrival. This is why a shipper confirms the collect arrangement with the consignee before booking: a slow or unreachable receiving party turns into a delayed delivery of cargo the shipper no longer controls.

Answer 2 describes freight prepaid, the opposite designation, under which charges are settled at or before shipment on the origin side. Answer 3 mistakes collect for open credit after delivery; the release of the cargo is precisely what awaits settlement, so payment does not trail delivery at the consignee's leisure. Answer 4 confuses the payment arrangement with the pricing of the goods; whether freight is built into the sale price is a matter for the sales contract and the agreed Incoterms 2020 rule, and it is separate from who pays the carrier and when.

The takeaway: prepaid and collect decide who pays and at which end, and the marking is verified, not assumed.

Fact-check appendix (audit notes — not for entry)

Definitions of freight rate, base freight, freight surcharge, ocean freight charge (OFC), air freight charge, bunker adjustment factor (BAF), currency adjustment factor (CAF), terminal handling charge (THC), documentation fee, chargeable weight, volumetric weight, weight or measurement (W/M), freight prepaid, freight collect, all-in freight rate and freight invoice follow the lesson brief 7.4 key terms. — source: lesson brief 7.4, Key Terms (curriculum database).

Ocean rates for containerised FCL cargo are commonly quoted per container; LCL cargo is commonly assessed on volume or weight as a share of the box. A forty-foot container is priced higher than a twenty-foot but typically well below double, being one unit of equipment lifted once; no amounts are taught. — source: research sheet trade-transport-standards.md, section 7; lesson briefs 7.3 and 5.3; the twenty-foot versus forty-foot pricing relationship is general industry practice.

Typical ocean freight-rate components (definitions, no amounts): ocean freight as the base line-haul rate; BAF as a fuel surcharge, now often expressed via attached index clauses; CAF as an exchange-rate surcharge; THC at origin (OTHC) and destination (DTHC) covering container handling at terminals; CFS charges for container freight station handling of LCL cargo; documentation fee (B/L issuance); seal charge; ISPS charge as security cost recovery under the SOLAS/ISPS Code. — source: research sheet trade-transport-standards.md, section 7.

GRI (general rate increase) is a liner-pricing term for an announced across-the-board upward adjustment of base freight rates, typically for contract seasons or capacity management on a trade lane; distinct from ad-hoc surcharges. — source: research sheet trade-transport-standards.md, section 7.

Liner services run published routes and schedules and sell space by the container; base rates emerge from published tariffs, contracts and market negotiation, with no universal price list. — source: research sheet trade-transport-standards.md, section 7 (liner vs tramp); lesson brief 7.1.

Peak-season and congestion surcharges are taught as concepts tied to recurring high-demand periods and port/terminal congestion; no amounts are quoted anywhere in the lesson. — source: research sheet trade-transport-standards.md, section 7 (rate components generally) and DO NOT CLAIM item 5 (no current rates, indices or carrier tariffs).

Air chargeable weight is the higher of gross weight and volumetric weight; the IATA volumetric formula is L x W x H in centimetres divided by 6,000. — source: research sheet trade-transport-standards.md, section 8.

The air waybill serves as receipt for the goods, evidence of the contract of carriage, freight bill and customs document, and is non-negotiable. — source: research sheet trade-transport-standards.md, section 4 (air waybill).

Under Incoterms 2020 the main carriage is arranged by the buyer under the E and F rules (EXW, FCA, FAS, FOB) and by the seller under the C and D rules (CFR, CIF, CPT, CIP, DAP, DPU, DDP). — source: research sheet trade-transport-standards.md, section 1.

Demurrage, detention and storage are separate time-based charge families that begin when free time ends; rates and free days are purely contractual tariffs with no standard international amounts; the detailed treatment belongs to Lesson 9.4. — source: research sheet trade-transport-standards.md, section 11 (general definitions only, one-sentence mention per the anti-repetition rule); WRITING_SPEC.md section 7.

The marking of shipments as freight prepaid or freight collect on transport documents, and the practice that destination local charges are commonly payable by the consignee in local currency, are general international business practice. — source: general business knowledge.

Revenue tonne as the unit of charge under W/M rating is standard industry usage. — source: general business knowledge, consistent with the W/M definition in lesson brief 7.4 key terms.

Cross-references are by name only, without re-teaching: Lesson 4.3 (Incoterms and Allocation of Responsibilities), Lesson 5.2 (Freight Quotations and Shipment Booking — quotation structure, validity; extended here into charge anatomy and scope rebuilding), Lesson 7.1 (Ocean Freight Operations and Vessel Schedules — liner context), Lesson 7.2 (Air Cargo Operations and Cargo Acceptance — chargeable weight and the AWB, referenced not re-derived), Lesson 7.3 (Containerisation — FCL/LCL, CFS, seals, TEU/FEU), Lesson 9.4 (Demurrage, Detention, and Storage Charges — one-sentence pointer). — source: WRITING_SPEC.md section 7; curriculum lesson briefs.

All quantities, rates and amounts in the calculation concept and the worked examples (1,150 / 180 / 150 / 60 / 15 / 25 / 220 / 190 US dollars; 1,575 / 2,040 / 2,080 / 2,040 US dollars; 80 x 60 x 50 centimetres, 26 kg, 40 kg, 4.00 / 0.80 / 0.40 US dollars per kilogram, 320 US dollars; and the smaller figures in the arithmetic concept) are illustrative for teaching purposes. — source: WRITING_SPEC.md rule 5.1; research sheet trade-transport-standards.md, DO NOT CLAIM item 5.

Module 7 Assessment — 12 questions

Enter via the instructor portal's Configure assessment step. Green ticks mark correct answers — do not enter them; enter options as-is.

Question 1 (tests lesson 7.1)

1

A truck delivering a laden container to the terminal at Tema reaches the gate forty minutes after the cargo cut-off for Thursday's sailing. The vessel is not expected alongside until Friday evening, and the driver argues that there is clearly time to load the box.

Which statement best explains the terminal's refusal?

1. The refusal is discretionary discipline for late trucks; with the vessel not yet alongside, the line can still add the container to the stowage plan at loading time.

2. The cut-off closes the sailing's cargo position because the stowage plan, the export and manifest data, and the verified weights are finalised from it, so a container the plan never contained cannot be absorbed and the booking rolls to the next sailing. ✓

3. A container may gate in after the cut-off provided its verified gross mass was declared on time, since the declared weight is the only planning data the vessel needs.

4. The cut-off exists so that the terminal can raise its invoices; once the shipper settles the terminal charges, the container can still make the sailing.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

A sailing is an engineered operation planned from its cut-off position, and the lesson names the three bodies of data that must be final before the plan can be built: the stowage plan, which decides where every container sits aboard and depends on verified weights; the export declaration and the vessel's manifest data, which must be complete before departure; and the VGM verification for every box. Once the cut-off passes, the plan is closed. A container that arrives afterwards is, in the lesson's words, a container the plan never accounted for, and the remedy is a rollover to the next sailing, about a week later on a weekly service.

Answer 1 treats the refusal as discretionary, but the gate is not queue discipline; it exists so that the plans can be built from facts, and a plan cannot be rebuilt around a box that gates in late. Answer 3 seizes the partial truth that VGM matters and ignores that the weights are one of three data sets that close with the cut-off, alongside stowage and manifest information. Answer 4 mistakes an operational planning deadline for a billing milestone; settlement does not rebuild a stowage plan.

The takeaway: the cargo cut-off is immovable because the whole sailing's loading plan is built from the cut-off position, so plan to beat it by a day.

Question 2 (tests lesson 7.1)

2

An exporter's container gated in at the terminal on Monday, ahead of Tuesday's cargo cut-off, with its verified gross mass declared. The shipping instruction was submitted complete the same day. On Thursday, after the documentation cut-off for the sailing has passed, the exporter asks the forwarder to change the consignee named on the bill of lading.

Which statement best describes the position?

1. No issue: with the container gated in and its VGM declared, every deadline for the sailing has been met, and the consignee can be updated at any time before the vessel reaches the port of discharge.

2. The bill of lading can never be amended after the documentation cut-off has passed, so the shipment must be cancelled and rebooked to change the consignee.

3. The documentation cut-off is the deadline for finalising the shipping instruction from which the carrier prepares the bill of lading; once it has passed, the details stand as submitted for the vessel's papers, and a change is an amendment to be arranged and confirmed with the carrier, not assumed. ✓

4. Consignee details are purely a destination customs matter, so the ocean cut-offs have no bearing on changing them.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The lesson teaches two separate clocks before a sailing. The cargo cut-off closes the terminal's gate against the packed container and its declared VGM; the documentation cut-off is the deadline for finalising the shipping instruction, the consignee, cargo description, weights, marks and container details from which the carrier prepares the bill of lading. Gate-in satisfies only the first clock. After the documentation cut-off, the details stand as submitted for the vessel's papers, so a consignee change is an amendment to be requested and confirmed with the carrier, subject to the carrier's procedures and costs, never a quiet edit assumed to have taken effect.

Answer 1 collapses the two deadlines into one and assumes a change can wait until arrival; the transport document is prepared, issued and used long before the vessel berths at destination. Answer 2 swings to the opposite extreme: the lesson says alterations after the cut-off are amendments, which are possible but must be arranged, not impossible. Answer 4 misplaces the responsibility; the consignee appears on the carrier's transport document, which is prepared at origin from the shipping instruction.

The takeaway: beating the cargo cut-off does not clear the paperwork; both cut-offs govern, and each sailing's actual dates are read from the booking confirmation.

Question 3 (tests lesson 7.1)

3

A trader in Lome has sold goods to a buyer in Rotterdam and books space on a liner service whose published rotation reads: Rotterdam, Antwerp, Tema, Lome, then back to Rotterdam.

Which statement correctly identifies the ports for this shipment?

1. The port of loading is Lome and the port of discharge is Rotterdam: on this service the vessel calls at Lome after Tema and then returns to Rotterdam, so cargo loaded at Lome is discharged at Rotterdam. ✓

2. The port of loading is Rotterdam and the port of discharge is Lome, because Rotterdam is listed first in the published rotation.

3. The service cannot carry the cargo, because a published rotation lists each port only once.

4. The port of loading must be Tema, because it is the larger container port and every service on the coast loads there first.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

A port rotation is the planned sequence of ports a vessel visits, read in calling order, and the service repeats its rotation voyage after voyage. On the leg that concerns this trader, the vessel sails from Europe, calls at Tema, then Lome, and returns to Rotterdam: the port of loading is Lome, where the cargo is loaded aboard, and the port of discharge is Rotterdam, where it is unloaded. Reading the rotation is also how a shipper confirms that a service can carry the cargo at all: both the origin and the destination must appear in the string, in the right order for the direction of the shipment.

Answer 2 reads the published list as a diary of the whole loop and forgets that the rotation repeats; the first port listed is not the loading port for every direction of trade. Answer 3 invents a rule; a rotation lists each port once per loop precisely because the service returns to it voyage after voyage. Answer 4 substitutes commercial folklore for the schedule: a port's prominence in the region has no bearing on which ports a named service calls, or in what order.

The takeaway: read the rotation in calling order for the direction of the cargo, and the port of loading and port of discharge identify themselves.

Question 4 (tests lesson 7.2)

4

A forwarder is quoting an air consignment of five identical cartons of canned food, booked as one consignment on the same flight. Each carton measures 50 x 40 x 30 centimetres and weighs 14 kilograms. A colleague who has heard that air freight is charged on volumetric weight quotes the consignment at 50 chargeable kilograms. All figures are illustrative for teaching purposes.

Which statement correctly determines the chargeable weight?

1. 50 chargeable kilograms is correct, because once carton dimensions are declared, air freight is always charged on volumetric weight.

2. The consignment is charged on 70 kilograms, because chargeable weight is the greater of the actual gross weight, 70 kilograms, and the volumetric weight, 5 multiplied by 50 x 40 x 30 divided by 6,000, which is 50 kilograms. ✓

3. The consignment is charged on 120 kilograms, because when the two weights differ, the volumetric weight and the actual weight are charged together.

4. The consignment pays a share of the whole unit load's chargeable weight, because once the cartons are built into an aircraft pallet they are charged as part of that unit load.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Chargeable weight is the greater of the actual gross weight and the volumetric weight, applied to the consignment's own figures. The volumetric weight of one carton is 50 x 40 x 30, which is 60,000 cubic centimetres, divided by the IATA divisor of 6,000: 10 kilograms. Five identical cartons give a total volumetric weight of 50 kilograms against a total gross weight of 70 kilograms. The actual weight is greater, so the consignment is charged on 70 chargeable kilograms. This is the mirror image of the light, bulky case the lesson dramatised: here the cargo exhausts the aircraft's payload before its space, so the scales, not the tape measure, decide the freight.

Answer 1 commits the inverse of the lesson's cardinal error: volumetric weight does not always govern; the rule is a comparison, and this time the actual weight wins. Answer 3 adds the two measures together, but they are alternatives in a comparison, never a sum; adding them would charge the same kilograms twice. Answer 4 merges the consignment into its unit load, but chargeable weight is assessed on each consignment's own weights, not on the consolidated unit load it later travels in.

The takeaway: compute both weights for the consignment, charge the greater, and never assume in advance which one will win.

Question 5 (tests lesson 7.2)

5

A forwarder delivers a consignment of general cargo to the airline's cargo terminal for a booked flight. The air waybill data and the booking state 12 pieces, but the physical count at acceptance finds 10 cartons.

Which statement best describes the position?

1. The airline must accept the consignment because the space was confirmed at booking; acceptance merely records what was delivered.

2. A two-piece discrepancy is minor: it can be noted on the documents and corrected with the consignee after arrival.

3. Checking the piece count is the forwarder's job before delivery; the airline or its ground handling agent checks only dangerous goods and security at the terminal.

4. Acceptance compares the physical consignment with the documents and the booking, and the three must agree, so the discrepancy stops acceptance until it is resolved, and the cargo does not fly until then. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

Cargo acceptance is the airline's gatekeeping process, performed by the airline or its ground handling agent at the terminal, and its decisive principle is that the goods, the documents and the booking must agree. The physical check counts the pieces against the document, so a consignment presenting ten cartons against twelve on the air waybill data and the booking is a discrepancy that stops the process until it is resolved: the documents are corrected or the missing pieces are found. Until then the consignment is not accepted, and if the fault is not fixed before the acceptance cut-off, the cargo misses the flight, whatever the departure board says.

Answer 1 treats a confirmed booking as a guarantee of acceptance, but the booking assures space on the flight; acceptance is a separate gate that the consignment itself must pass. Answer 2 waves the discrepancy through, yet goods and a document that contradicts them are exactly what acceptance exists to stop, because the paperwork travels with the cargo into customs at the destination. Answer 3 hands the count to the forwarder, but the count at acceptance is the airline-side comparison of the physical goods against the document.

The takeaway: at acceptance the consignment is measured against its papers, and any disagreement stops the process until goods and papers agree.

Question 6 (tests lesson 7.2)

6

A new exporter delivers a consignment of leather goods to the air cargo terminal for a booked international flight. At the terminal, the exporter's colleague says the cargo need not be screened because it is ordinary general cargo.

Which statement best describes the security requirement?

1. Every consignment's security status must be established before loading: the cargo either travels under an approved trusted shipper's assessed status, commonly described as a known consignor, or it is screened before acceptance into the air side, and the status cannot be claimed at the desk. ✓

2. Screening applies only to dangerous goods, because ordinary general cargo presents no threat to an aircraft.

3. A shipper may declare its own trusted-shipper status at the terminal on the day of travel and travel without screening from that point.

4. Consignments on dedicated freighter aircraft are exempt from screening, because no passengers are carried on the flight.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

Air cargo is security cargo: no consignment may be loaded unless its security status is established. The lesson teaches two broad routes, described generically because each country implements them through its own national regime. Cargo from an approved trusted shipper, commonly described as a known consignor, enters the air side on the strength of that shipper's assessed supply-chain controls, and the approval is conditional on maintaining the required procedures. Every other consignment must be screened, by scanning equipment or physical examination, before or at acceptance into the air side, and the status travels with the cargo as data and paperwork. Cargo whose status cannot be established is not loaded; it waits, which in practice can mean a missed flight.

Answer 2 narrows security to dangerous goods, but screening examines cargo for prohibited articles whatever the commodity. Answer 3 converts an assessed approval into a self-declaration, which is precisely what the concept does not allow: trusted-shipper status is earned

through sustained compliant procedures, not asserted at the desk. Answer 4 invents a freighter exemption; the security concern is the loaded aircraft and the air cargo chain, not the passenger cabin alone.

The takeaway: security status is part of the acceptance picture, so allow time for it before the cut-off and never assume it can be claimed on the day.

Question 7 (tests lesson 7.3)

7

A project forwarder in Tema must ship an excavator that is wider than a standard container's internal width, together with fourteen cartons of spare parts of ordinary general cargo.

Which equipment plan is correct?

1. A high-cube container, because the extra internal height creates room for tall machinery.

2. An open-top container, because machinery is lowered in from above through the removable roof, which also solves the width problem.

3. A flat-rack container, on which the out-of-gauge excavator is lashed and secured, with the cartons of spare parts packed separately in an ordinary dry container. ✓

4. One forty-foot dry container for everything, because cargo securing straps can restrain the excavator safely between the cartons.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The lesson's selection rule turns on the cargo's own dimensions. An excavator wider than a standard container's internal width is out-of-gauge cargo, and out-of-gauge cargo points to a flat rack: a platform with fixed or collapsible end structures but no sides and no roof, built for items that are wider, longer or simply not rectangular, with lashing, securing and weather protection planned as part of the loading. The spare parts are ordinary general cargo with no special requirement, so they take a dry container beside the flat rack.

Answer 2 is the attractive trap, and it fails on one word: width. An open-top has a removable roof for top loading, which answers cargo that is too tall for the door aperture or too awkward to load through doors, but its sides are fixed, so cargo wider than the box cannot sit inside it. Answer 1 fails because a high cube adds height on the same footprint; the internal width is unchanged, and no amount of extra height admits a load that is too wide. Answer 4 ignores both

the width and the lesson's compatibility rule: heavy machinery braced among fragile cartons is an inadvisable mix even if it physically fit.

The takeaway: characterise the cargo first, then choose the box; top loading answers height, and only a flat rack answers width.

Question 8 (tests lesson 7.3)

8

A small exporter in Takoradi ships about three pallets, roughly 8 cubic metres and 900 kilograms, to a buyer in Copenhagen four or five times a year. A colleague insists that every ocean shipment should be booked as FCL because LCL cargo travels loose in the ship's hold and is always damaged.

Which statement gives the correct guidance?

1. The colleague is right: FCL is always cheaper than LCL, whatever the volume, because a whole container is one price negotiation.

2. At this volume LCL is the natural choice: the consignment occupies part of a container consolidated with other shippers' cargo at a container freight station, the consolidated box travels sealed like any container, and the choice should weigh cost per unit, handling risk, timing and cargo sensitivity rather than follow a rule that FCL is always better. ✓

3. LCL cargo cannot be sealed or tracked, so the exporter must book a twenty-foot container even though it would travel nearly empty.

4. Volume is irrelevant to the decision; the exporter should simply alternate between FCL and LCL from one shipment to the next.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

The lesson's first decision factor is volume against a full box: a twenty-foot container offers about 33 cubic metres, and three pallets of roughly 8 cubic metres use only a fraction of it, which makes the consignment a natural LCL. An LCL shipment occupies part of a container and is consolidated with other shippers' cargo at a container freight station, and the consolidated box is sealed and moves as one unit, exactly like any container. The choice should then weigh the remaining factors: cost per unit, handling risk at stuffing and deconsolidation, the earlier CFS cut-off, and cargo sensitivity.

Answer 1 asserts that FCL is always cheaper; the lesson teaches the opposite shape: at low volumes the LCL share usually costs less in total, and the crossing point arrives well before a box is full, so the exporter would pay for space it does not use. Answer 3 describes LCL cargo as unsealable, but the seal travels on the consolidated container's doors, not on individual consignments, and the container number identifies the box wherever it moves. Answer 4 replaces the decision with a habit; the five factors exist because the shipment, not a formula, decides.

The takeaway: neither FCL nor LCL is universally right; small, occasional volumes are LCL's natural ground, packed well for the neighbours a shared box brings.

Question 9 (tests lesson 7.3)

9

A booking confirmation covers one forty-foot high-cube container and two twenty-foot dry containers on the same sailing.

Which statement correctly states the capacity the booking consumes in the industry's counting units?

1. Three TEU, one for each physical box on the booking.

2. Five TEU, because the high cube's extra height earns it three TEU against one for each twenty-foot box.

3. It cannot be stated until sailing, because TEU counts the cargo actually carried rather than the boxes booked.

4. Four TEU, because the high cube counts as one FEU, generally equal to two TEU, and each twenty-foot box counts as one TEU. ✓

Correct answer: Answer 4

Explanation

The correct answer is Answer 4.

The twenty-foot equivalent unit measures nominal length: one TEU is one nominal twenty-foot container, and one FEU is one nominal forty-foot container, generally equal to two TEUs. The booking covers one forty-foot box and two twenty-foot boxes: the forty-foot high cube counts as one FEU, that is two TEU, and each twenty-foot box counts as one TEU, giving four TEU in total. The high cube's extra height changes nothing, because the unit measures the nominal length, not the volume; the lesson makes this explicit when it notes that a forty-foot high cube still counts as one FEU despite standing a foot taller than the standard box.

Answer 1 counts physical boxes, which confuses the number of units handled with the capacity measure in which vessels, ports and the industry count their business. Answer 2 awards the high cube a third unit for its extra height, but the TEU has no volumetric exchange rate; the high cube's roughly 75 to 76 cubic metres change the cargo plan, not the count. Answer 3 ties the count to the fill, yet TEU and FEU measure container capacity, not the cargo actually carried; an empty forty-foot box is still one FEU.

The takeaway: count TEUs from the nominal length of the boxes booked, and let a high cube's height influence the load plan, never the arithmetic.

Question 10 (tests lesson 7.4)

10

A forwarder in Accra is rating an LCL consignment of six crates of machine spare parts for ocean shipment. The crates weigh 9,000 kilograms in total and measure 8 cubic metres. The LCL rate is quoted per weight or measurement tonne, whichever produces the higher charge. All figures are illustrative for teaching purposes.

On which quantity will the freight be charged?

1. On 8 revenue tonnes, because LCL cargo shares a box and is therefore always charged on the space it occupies.

2. On 9 revenue tonnes, because the chargeable quantity under weight or measurement rating is the greater of the weight in metric tonnes and the volume in cubic metres, and dense cargo exhausts the weight allowance first. ✓

3. On 17 revenue tonnes, because the weight and the measurement are added together before the rate is applied.

4. On the number of crates, because consolidated cargo is charged by the handling unit rather than by weight or measurement.

Correct answer: Answer 2

Explanation

The correct answer is Answer 2.

Weight or measurement rating exists because a consignment can exhaust either the vessel's weight allowance or its space, so the tariff charges whichever measure produces the higher quantity: weight in metric tonnes against measurement in cubic metres, the greater becoming the revenue tonnes charged. Six crates weighing 9 tonnes and measuring 8 cubic metres convert to 9 against 8, so the weight wins and the consignment is charged on 9 revenue tonnes. This is the dense-cargo mirror image of the lesson's worked case, in which light plastic household

goods paid on 6.8 cubic metres rather than 2.4 tonnes; the method is identical, only the winner changes with the cargo.

Answer 1 assumes LCL is always charged on space, which would undercharge exactly the dense cargo that fills a box's weight allowance long before its volume; weight governs only when it is the greater figure, as it is here. Answer 3 adds the two measures, but they are alternatives in a comparison, never combined; adding them would charge part of the consignment twice. Answer 4 invents a per-package basis, while consolidation rates are built on the consignment's weight or measurement.

The takeaway: under W/M, convert the consignment to both tonnes and cubic metres, charge the higher, and let the figures, not the cargo's appearance, choose the basis.

Question 11 (tests lesson 7.4)

11

An importer in Kumasi is arranging shipment of machinery from Shanghai to Tema. Forwarder P quotes 2,300 US dollars, port to port, and lists destination terminal handling, assistance with customs clearance, and delivery to the warehouse as excluded charges that together add 480 US dollars, payable locally at destination. Forwarder Q quotes 2,700 US dollars, door to door, including all of those charges. A colleague says Forwarder P is 400 US dollars cheaper and should be chosen immediately. All figures are illustrative for teaching purposes.

Which statement is correct?

1. Forwarder P is cheaper, and the 400 US dollar difference stands, because destination charges are minor extras that never change a comparison.

2. The two quotations cannot be compared at all, because every forwarder builds its prices differently and no common basis exists between offers.

3. Rebuilt to equal scope, Forwarder P's offer costs 2,780 US dollars, so Forwarder Q is 80 US dollars cheaper on the same scope; headline totals built on different scopes cannot be compared directly. ✓

4. Forwarder Q should be chosen automatically, because a door-to-door quotation is always better value than a port-to-port quotation regardless of the figures.

Correct answer: Answer 3

Explanation

The correct answer is Answer 3.

The two quotations are built on different scopes, so the headline comparison measures scope, not value. Rebuilt to equal scope, Forwarder P's offer is 2,300 plus the 480 US dollars of destination charges it excludes, which is 2,780; Forwarder Q's 2,700 already covers the door. On equal scope Q is 80 US dollars cheaper, and the colleague's 400-dollar saving reverses once the destination tail is priced. The lesson's reading method is exactly this rebuild: read each quotation down to its lines, base freight, surcharges and origin local charges, add the destination charges it excludes, and compare only totals that cover the same scope, before weighing conditions such as validity.

Answer 1 repeats the headline error and pairs it with the lesson's most expensive assumption, that destination charges are minor extras; the local tail can rival the main carriage itself. Answer 2 abandons comparison altogether and forfeits the line-by-line discipline that makes quotations readable. Answer 4 buys inclusiveness as such, yet door to door is not automatically value; Q wins here on the rebuilt figures, not on its scope label.

The takeaway: rebuild both quotations to the same scope, then compare price, then conditions; a low headline measures what is missing, not what is saved.

Question 12 (tests lesson 7.4)

12

An exporter in Tema has sold a consignment of cocoa products to a buyer in Dakar on terms under which the buyer, not the exporter, arranges the main carriage (an E or F rule of Incoterms 2020). The forwarder asks how the freight payment should be designated before booking.

Which designation and guidance is correct?

1. Freight collect: the charges are payable at destination by the designated receiving party, so before booking the exporter should confirm that the buyer can and will settle promptly, because carriers release cargo against payment. ✓

2. Freight prepaid, because the goods leave the seller's country and the seller therefore always settles the freight before shipment.

3. Either designation, because carriers treat prepaid and collect identically and the marking carries no operational effect.

4. Freight collect, with the buyer free to take delivery on arrival and settle the freight afterwards at its convenience.

Correct answer: Answer 1

Explanation

The correct answer is Answer 1.

A freight collect designation indicates that the charges are payable at destination, or by the designated receiving party, under the agreed arrangement. The lesson links the marking to the sales terms: under Incoterms 2020 the E and F rules place the arrangement of the main carriage with the buyer, and where the buyer arranges the carriage the movement is commonly booked on collect terms so the carrier recovers its charges at the far end. Because carriers release cargo against settlement, the exporter should confirm before booking that the buyer can and will pay promptly, since a slow or unreachable consignee strands the cargo at destination, out of the exporter's control.

Answer 2 flips the logic: freight prepaid marks charges settled at or before shipment, which follows the seller arranging the carriage under the C and D rules, not this sale. Answer 3 declares the marking a formality, but prepaid and collect decide who pays and at which end, and the marking on the document, not the parties' assumption, is the authority. Answer 4 detaches payment from delivery, yet release of the cargo is precisely what awaits settlement; collect is not open credit after arrival.

The takeaway: prepaid or collect follows the party that arranges the main carriage, is verified before booking, and is marked consistently on the documents.